From Semantics to Paths: Carrier-Aware Global Emergency Routing via Parallel Road Network and Land-Cover Inference

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***Abstract*—Effective emergency routing in disaster-stricken areas is a critical challenge, particularly when conventional digital maps become obsolete due to infrastructure damage. Existing solutions often treat road extraction and land-cover classification as isolated tasks, failing to provide a holistic navigation solution for diverse rescue carriers. To address this, we propose a novel carrier-aware global emergency routing framework that transforms a single remote sensing image into navigable paths. We first construct two large-scale datasets tailored for the complex topography of the Sichuan-Tibet region: STRoadSet for road extraction and STLandSet for land-cover classification. We then introduce a parallel semantic inference mechanism comprising HC-RoadNet, a connectivity-preserving road extraction network, and NX-SegNet, a ConvNeXt-based land-cover classification network. To convert the raw, high-recall road mask into a planning-ready layer, we further design a four-step refinement pipeline that combines span-based filtering, KD-Tree-indexed fracture repair, land-cover-prior tunnel recovery, and morphological bridging. The refined road layer is fused with the land-cover map into a unified, carrier-aware cost map, and an improved A* search with a turn penalty and a road-priority hierarchical strategy computes routes under three optimization modes (time, energy, and balanced). Across diverse carrier profiles and optimization objectives, experimental results confirm that the proposed framework is effective and computationally efficient, meeting the time-critical demands of emergency navigation. The full-source code and datasets are publicly available at: https://github.com/dongxr2/RS-Routing/**

***Index Terms*—Carrier-aware path planning, deep learning, emergency routing, remote sensing, semantic segmentation, topology preservation.**

I. Introduction

EFFECTIVE and timely global routing is a pivotal factor for the success of emergency response missions, significantly influencing outcomes in domains, such as search and rescue, disaster relief, and logistics support [1]. In the aftermath of large-scale natural disasters, conventional navigation systems—which predominantly depend on static, pre-existing digital road networks—often encounter substantial operational limitations [2], [3]. Prominent events, including the 2008 Wenchuan earthquake [4] and the 2021 Henan floods [5], have demonstrated the vulnerability of transportation infrastructure, where critical routes can be severely disrupted or destroyed, rendering conventional digital maps rapidly outdated. Under such volatile and high-risk conditions, rescue teams are frequently left without dependable navigational information, resulting in reliance on inefficient and hazardous exploratory movements guided primarily by remote sensing imagery [6], [7]. This issue is especially pronounced in remote and disaster-prone regions, such as the Sichuan-Tibet area in China, which are not only subject to recurrent geological hazards but also characterized by extensive data-scarce zones with incomplete or nonexistent road network documentation [4], [8], [9], [10]. Therefore, developing effective solutions for these scenarios is of considerable importance.

Remote sensing imagery, offering both frequent updates and extensive spatial coverage, has become a highly viable data source for achieving timely situational awareness in disaster-affected regions [11]. Although numerous studies have focused on the intelligent interpretation of remote sensing imagery, research dedicated to applying such data to emergency navigation routing remains relatively limited. Current efforts are mainly divided into two parallel semantic perception tasks: road network extraction [12], aimed at identifying potential pathways, and land-cover classification [13], [14], which characterizes the spatial distribution of surface types. However, relying solely on road network extraction presents a significant challenge, as the accurate identification of highly connected networks is notoriously difficult and often imperfect. This can lead to fragmented road segments in the output due to dataset limitations or insufficient model accuracy, creating critical discontinuities in potential routes. To address this,

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our work posits that integrating land-cover information is essential. Land cover is not a direct measurement of terrain elevation or slope; rather, it serves as a proxy for traversability impedance. For instance, a forest pixel implies vegetation-induced friction for wheeled vehicles, while a water-body pixel constitutes a hard barrier for ground carriers but a frictionless corridor for amphibious platforms. This semantic abstraction allows land-cover information to encode carrier-specific mobility constraints that are not derivable from a road map alone. By synthesizing the extracted road network with a traversability map derived from land-cover classification, we can construct a unified navigation graph. This approach not only supports tactical off-road navigation in emergencies but also effectively mitigates the issue of network fragmentation by providing alternative paths across gaps, thus enhancing the robustness of the routing solution.

Existing benchmark datasets for road extraction and land-cover classification, while valuable for advancing computer vision techniques, exhibit significant limitations when applied to emergency routing scenarios. Popular datasets, such as DeepGlobe Road Extraction [15], SpaceNet [16], and the Massachusetts Roads Dataset [17], predominantly focus on well-developed urban and suburban areas with clear road structures and favorable imaging conditions. Similarly, widely used land-cover datasets like ISPRS Potsdam and Vaihingen [18] are constrained to small-scale European urban scenes with high-resolution imagery and comprehensive ground-truth (GT) annotations. These datasets, by design, prioritize environments with well-maintained infrastructure and minimal damage, which stands in stark contrast to the chaotic, rapidly changing conditions characteristic of disaster zones. Moreover, most existing datasets address road extraction and land-cover classification as isolated tasks, lacking the integrated annotations necessary for holistic emergency navigation planning. The geographic homogeneity of these datasets—predominantly covering developed regions in North America and Europe—further limits their applicability to diverse disaster-prone areas, such as mountainous regions in Southwest China, where complex topography, sparse infrastructure, and frequent geological hazards present fundamentally different challenges.

Recent progress in artificial intelligence has provided technical support for remote sensing-based semantic perception. Models, such as U-Net [19], with its encoder–decoder structure and skip connections, have become a de facto standard for road segmentation tasks, demonstrating strong performance in capturing fine-grained spatial details. DeepLabV3 [20] introduced atrous convolution and spatial pyramid pooling modules, enabling multiscale feature extraction that proves particularly effective for land-cover classification across diverse terrain types. More recently, Transformer-based models, such as Vision Transformer [21] and Swin Transformer [22], have demonstrated superior capability in capturing long-range dependencies and global context, which is crucial for understanding connectivity patterns in fragmented road networks. Hybrid architectures combining convolutional and attention mechanisms have further pushed the boundaries of semantic segmentation accuracy in remote sensing applications [23]. Beyond single-task models, multitask learning frameworks have emerged as promising approaches for optimizing multiple related objectives, enabling efficient feature sharing and improved generalization [24], [25], [26]. However, despite these architectural advances, most existing methods treat road extraction and land-cover classification as independent problems, failing to leverage their inherent semantic correlations at the decision level. Furthermore, the adaptation of these models to disaster scenarios, where image quality may be degraded, infrastructure partially destroyed, and terrain characteristics drastically altered, remains underexplored, highlighting a critical gap between technical capability and practical deployment in emergency contexts.

Beyond semantic perception, path planning constitutes the second indispensable pillar of emergency routing. As a mature theoretical foundation, graph-search algorithms, such as Dijkstra's and A*, have long served as the algorithmic backbone of geographic information system (GIS)-based routing systems [27], [28]. Within the disaster-response context specifically, however, research has branched into several specialized directions that move beyond generic shortest path computation. Evacuation and egress routing optimizes the movement of affected populations out of hazard zones under capacity, congestion, and time-window constraints [29]. Humanitarian logistics and relief-distribution routing addresses last-mile delivery and emergency vehicle dispatch over partially damaged transportation networks [30]. Robust and stochastic shortest path formulations explicitly model edge failures and travel-time uncertainty induced by infrastructure disruption, seeking routes that remain feasible under a range of postdisaster network states [31]. Despite their diversity, these works share a common premise: the underlying graph topology is assumed to be known a priori, derived either from pre-existing road databases or from manually surveyed networks. Consequently, an important aspect remains underexplored. None of these approaches addresses the upstream problem that is central to our setting—recovering a routable graph directly from a single postdisaster remote-sensing image when conventional digital maps have become obsolete—and none models the true physical traversal cost for heterogeneous rescue carriers over unstructured terrain, given that the 2-D semantic content of remote sensing imagery does not directly reveal such cost. This twofold deficiency motivates a tighter coupling between semantic perception and carrier-aware cost modeling in postdisaster conditions, which is the gap our framework is designed to fill.

Despite the aforementioned advances in datasets and methodologies, significant gaps remain in developing integrated solutions for emergency routing that can address the unique challenges of disaster scenarios. Specifically, there is a lack of: 1) specialized datasets capturing the environmental complexity and infrastructure vulnerability characteristic of disaster-prone regions; 2) unified frameworks that integrate road network and land-cover information for robust route planning; and 3) carrier-aware routing algorithms that can adapt to different vehicle types and operational constraints in emergency contexts. These limitations motivate the present work.

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In a previous study [32], we introduced a multisource remote sensing data fusion framework capable of integrating diverse imagery and datasets into a unified representation, thereby laying a data-level groundwork for subsequent applications. Building on these advancements, this article proposes a novel framework that enables carrier-aware global emergency routing using only a single-remote sensing image. First, we introduce parallel semantic inference, employing two independently trained branch models to implement road network extraction and land-cover identification, respectively. The dual-branch design is intended to enhance the fault tolerance and maintainability of the process; a failure or domain shift in one branch does not corrupt the other, and it allows for independent retraining for specific geographic regions. However, the accurate extraction of highly connected road networks remains a persistent challenge, often resulting in network discontinuities due to dataset issues or limited extraction accuracy. To bridge the gap between raw semantic masks and a routable graph, we further introduce a multistage refinement pipeline. This pipeline comprises span-based connected-component filtering, endpoint-based fracture repair via KD-Tree spatial indexing, land-cover-prior tunnel recognition, and morphological dilation bridging. Specifically, the tunnel recognition process utilizes land-cover priors to repair the road network, thereby implementing a loosely coupled form of joint reasoning between the land-cover and road networks. This pipeline systematically restores the topological connectivity compromised by the high-recall, wide-boundary characteristic of the road extraction network. The refined road layer and the land-cover layer are then integrated into a unified, carrier-aware cost map, parameterized according to factors such as carrier type, traversal speed, and energy consumption. This synthesis is critical, as it allows the traversable ground surface to serve as a bridge across gaps in the extracted road network, thereby ensuring route integrity. Finally, an improved A* search algorithm augmented with a turn penalty and a road-priority hierarchical strategy is employed to generate optimal or near-optimal routes that effectively balance the high-speed travel on roads with tactical off-road navigation.

The main contributions of this work are summarized as follows.

1) To address the problem that existing benchmark datasets focus on developed urban scenes in North America and Europe, treat road extraction and land-cover classification as isolated tasks, and lack annotations for the complex terrain of disaster-prone regions, we construct two large-scale semantic segmentation datasets dedicated to the topographically complex and disaster-prone Sichuan-Tibet region. Specifically, STRoadSet contains 30 687 image-label pairs for road extraction, and STLandSet contains 1802 pairs for land-cover classification.
2) To address the problem that conventional road-segmentation networks optimize for pixelwise accuracy and consequently produce fragmented road masks that break the connectivity of potential routes, we propose HC-RoadNet, a road extraction network that combines an EfficientNet-B7 encoder with a heterogeneous attention decoder and a topology-preserving composite loss, explicitly trading marginal precision for substantial gains in connectivity. We argue that this asymmetric tradeoff is essential in postdisaster contexts.
3) To address the problem that general-purpose land-cover networks optimize only for overall accuracy and overlook the balanced, per-class impedance estimates required by the downstream cost-map fusion, we propose NX-SegNet, a land-cover classification network that couples a ConvNeXt-Tiny encoder with a coordinate attention (CA) bridge and an attention-guided decoder, achieving balanced precision and recall across multiple nonstructured land-cover classes.
4) To address the problem that prior methods enforce topological constraints within the segmentation loss, sacrificing modularity and requiring model retraining, we design a multistage road network refinement pipeline that converts the raw, high-recall road mask into a topologically clean and routable layer, improving the accuracy and connectivity of the road network without requiring model retraining.
5) To address the problem that classical GIS planners such as A* and Dijkstra assume terrain isotropy and cannot capture the true physical traversal cost of heterogeneous rescue carriers, we formulate a carrier-aware unified cost map and an improved A* search with a turn penalty and road-priority hierarchical decomposition. We explicitly model the traversal costs for three types of emergency carriers (logistics vehicle, amphibious hovercraft, and on-foot rescue team) and experimentally validate the effectiveness of the proposed method under three optimization modes (time-priority, energy-priority, and balanced).

The remainder of this article is organized as follows. Section II introduces the study area and details the two newly constructed datasets, STRoadSet and STLandSet, developed for this research. Section III elaborates on the proposed global emergency routing methodology, including the parallel semantic inference framework, the road network refinement pipeline, the carrier-aware cost-map generation, and the improved A* planning algorithm. Section IV presents the experimental setup, results, and analysis for road extraction, land-cover classification, the refinement pipeline, and global routing, with a quantitative path-level comparison between inferred and GT road networks. Section V discusses the limitations of the current work and suggests future research directions. Finally, Section VI concludes this article.

II. Dataset Development

To support the development and validation of our emergency routing framework, we constructed two large-scale datasets centered on a representative disaster-prone region in the Sichuan-Tibet area of China, geographically bounded by the corner coordinates (103.4241°E, 29.8355°N) and (102.8993°E, 30.1735°N). This region is characterized by its intricate topography, diverse land-cover, and high susceptibility to geological hazards, making it an ideal testbed for our proposed framework. Building upon our foundational data fusion work in [32],

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(a) (b)

Fig. 1. Overview of the STRoadSet dataset for road network extraction. (a) Mosaicked remote sensing imagery of the study area sourced from Tianditu (zoom level 17). (b) Corresponding GT binary road network mask, where black pixels denote the road class and white pixels denote the nonroad background.

we constructed two distinct, large-scale datasets: STRoadSet for road network extraction and STLandSet for land-cover classification. These datasets serve as the empirical basis for training and evaluating the parallel semantic inference models central to our methodology.

A. STRoadSet for Road Network Extraction

To train and evaluate models for robust road network extraction in complex disaster-prone environments, we developed the STRoadSet dataset. This dataset is built upon high-resolution satellite imagery (zoom level 17, approximately 1.2 m/pixel) sourced from Tianditu, China's authoritative national platform for geospatial information services [33].

STRoadSet comprises 30 687 image-label pairs, created by systematically tiling the remote sensing imagery of the study area. Each pair consists of a 256 $\times$ 256 pixel satellite image and a corresponding pixel-aligned binary road mask. The GT road network labels were generated by rasterizing the official vector road data provided by the Tianditu platform, ensuring high positional accuracy and topological integrity. As illustrated in Fig. 1, the dataset provides the input remote sensing image [see Fig. 1(a)] and its corresponding label [see Fig. 1(b)]. The label is a binary mask, where black pixels represent the road class, and white pixels represent the nonroad background. The dataset exhibits the severe foreground–background imbalance characteristic of road extraction: road pixels constitute only 1.66% of the total, with the remaining 98.34% being nonroad background. For the purposes of model development and evaluation, the entire dataset was partitioned into training, validation, and test sets following a standard 8:1:1 ratio. This dataset is specifically designed to address the challenges of road extraction in topographically complex regions, which are characteristic of the emergency scenarios targeted in this research.

B. STLandSet for Land-Cover Classification

To facilitate carrier-aware off-road navigation, a second dataset, STLandSet, was developed for detailed land-cover classification and terrain characterization. STLandSet integrates the same Tianditu satellite imagery used in STRoadSet with semantic labels adapted from the GlobeLand30 dataset, a 30-m resolution global land-cover data product renowned for its high accuracy and comprehensive coverage, which delineates ten primary land-cover types: Cultivated Land, Forest, Grassland, Shrubland, Wetland, Water Bodies, Tundra, Artificial Surfaces, Bareland, and Permanent Snow and Ice [32], [34].

A geographical analysis of our specific study area confirmed the absence of five of these classes within the bounding box, namely, Grassland, Shrubland, Tundra, Bareland, and Permanent Snow and Ice. Consequently, the STLandSet training data effectively contains the five remaining classes relevant to the region. Crucially, however, NX-SegNet (detailed in Section III-C) was deliberately designed to operate on the full ten-class label space rather than the regionally observed five-class subset. This design decision serves two purposes: 1) it preserves architectural consistency with the established GlobeLand30 standard, ensuring transferability to other geographic regions where the missing classes may occur and 2) it allows the carrier-aware cost map (see Section III-D) to retain parameter slots for the absent classes, enabling out-of-the-box deployment in a different study area without retraining the segmentation head.

To establish the initial spatial alignment between the high-resolution Tianditu imagery and the low-resolution GlobeLand30 labels, we performed a manual georeferencing process in Global Mapper, as illustrated in Fig. 2. Ten ground control points (GCPs) were manually selected at salient geomorphological features readily identifiable in both data sources—primarily road–river intersections, distinctive river

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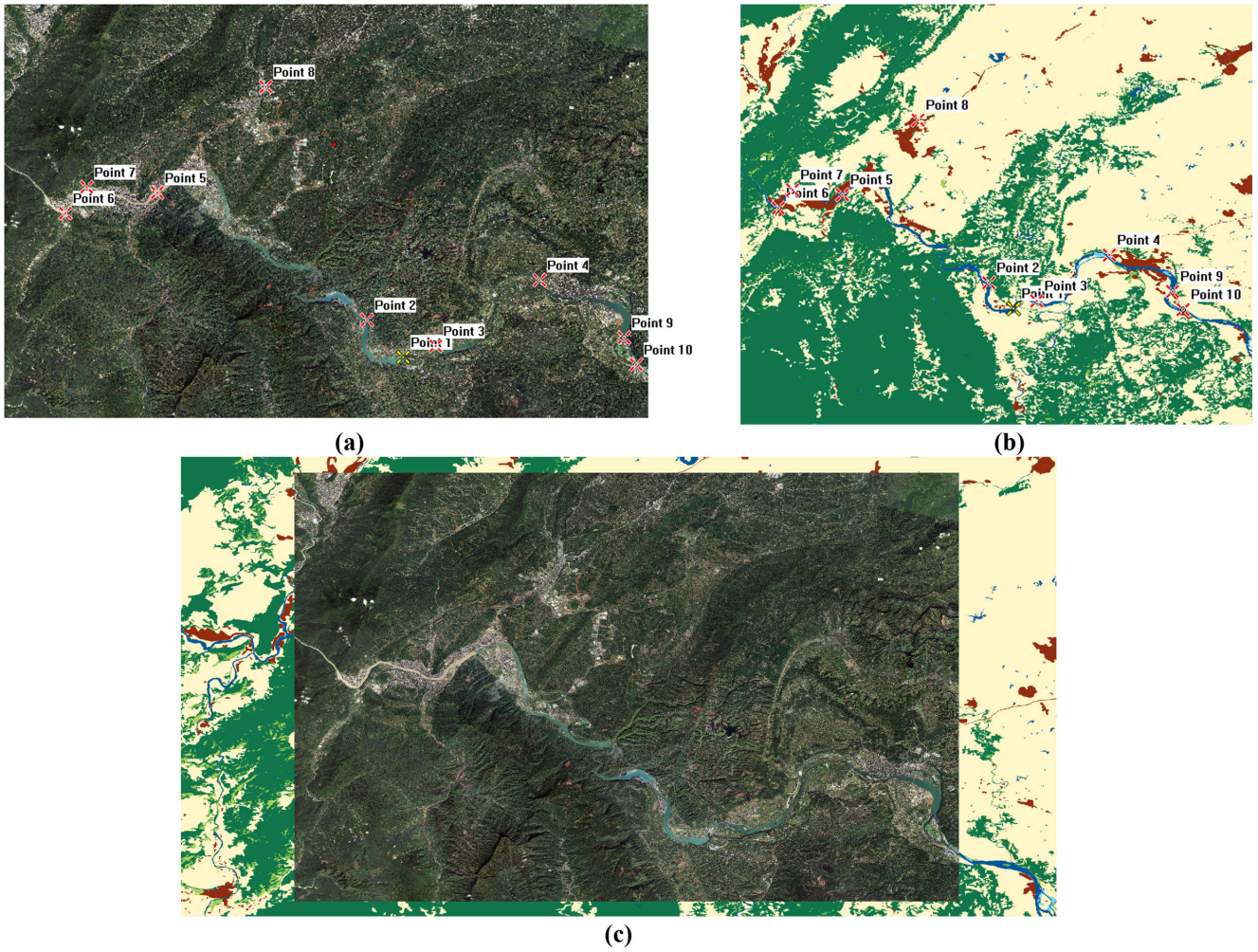

Fig. 2. Workflow for pixel-level co-registration of multisource geospatial data. (a) and (b) Ten GCPs selected on the Tianditu satellite imagery and the GlobeLand30 land-cover layer, respectively. (c) Final registered output with accurate pixel-level correspondence.

bends, and isolated artificial-surface patches—while ensuring an approximately uniform spatial distribution across the study area to avoid registration bias toward any subregion. Fig. 2(a) and (b) shows the selected GCPs on the satellite imagery and the GlobeLand30 layer, respectively, while Fig. 2(c) displays the registered output with pixel-level correspondence.

Following georeferencing, the land-cover distribution of the target area was obtained, as shown in Fig. 3. Subsequently, an automated processing pipeline was implemented to refine the alignment and generate training-ready data. A key challenge arose from the resolution disparity between the two sources. Directly upscaling the low-resolution GlobeLand30 labels to match the high-resolution satellite imagery introduces color-interpolation artifacts, manifesting as pixels with nonstandard RGB values along class boundaries. To rectify this, we performed a boundary-artifact cleaning step. The algorithm first identifies all pixels whose RGB values do not match any of the five predefined standard class colors. For each such artifact pixel, an expanding spiral neighborhood search (from a 3×3 window outward) is conducted, and the color of the first standard-class pixel encountered is assigned to the artifact pixel, effectively reassigning it to the nearest valid land-cover type while preserving crisp categorical boundaries.

After this cleaning process, the refined low-resolution label was upscaled by a factor of eight using nearest neighbor resampling to preserve categorical edges, while the original satellite imagery was resized using bilinear resampling, which is better suited for continuous-tone imagery. The resulting aligned image–label pairs were then cropped into 256×256 pixel patches; for each patch, we generated both an RGB satellite image and a three-channel RGB label encoding the class index. This entire pipeline yielded a final dataset of 1802 image–label pairs, providing the essential semantic layer for constructing the traversability cost map used in our global routing algorithm. Following the same protocol as STRoadSet, STLandSet was split into training, validation, and testing sets at an 8:1:1 ratio. The class distribution is highly imbalanced: cultivated land (53.18%), and forest (41.51%) dominate, followed by artificial surface (3.34%), water body (1.82%), and wetland (0.15%), while the remaining five classes are absent. This imbalance motivates the macro-averaged metrics and mean intersection over union (mIoU) in Section IV-C1, which weight all classes equally.

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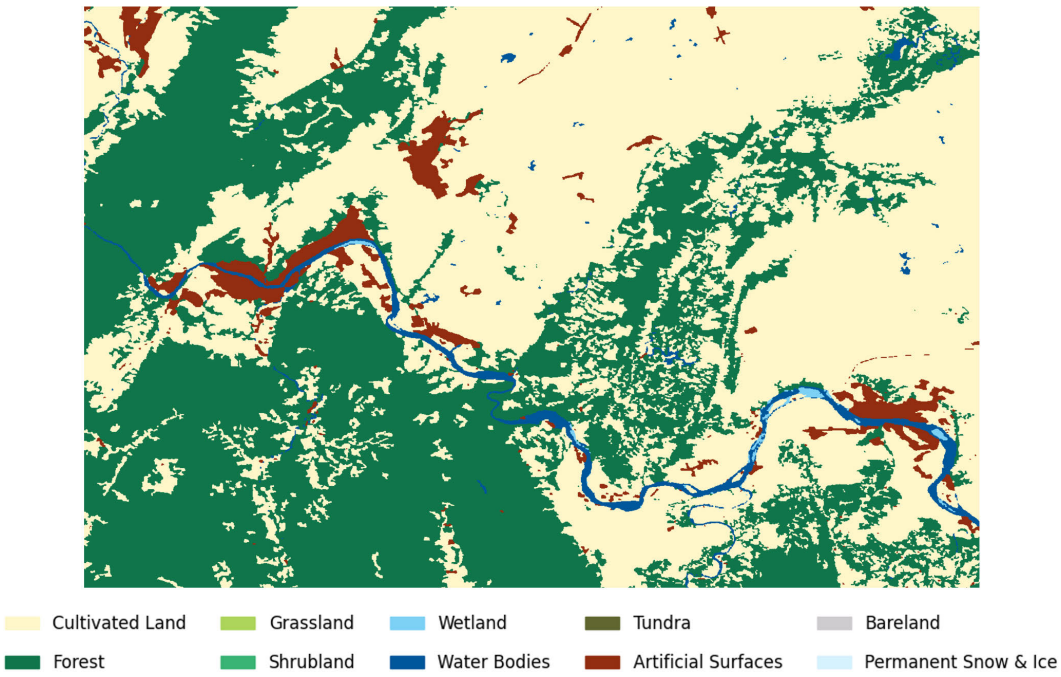

Fig. 3. Georeferenced land-cover map of the study area.

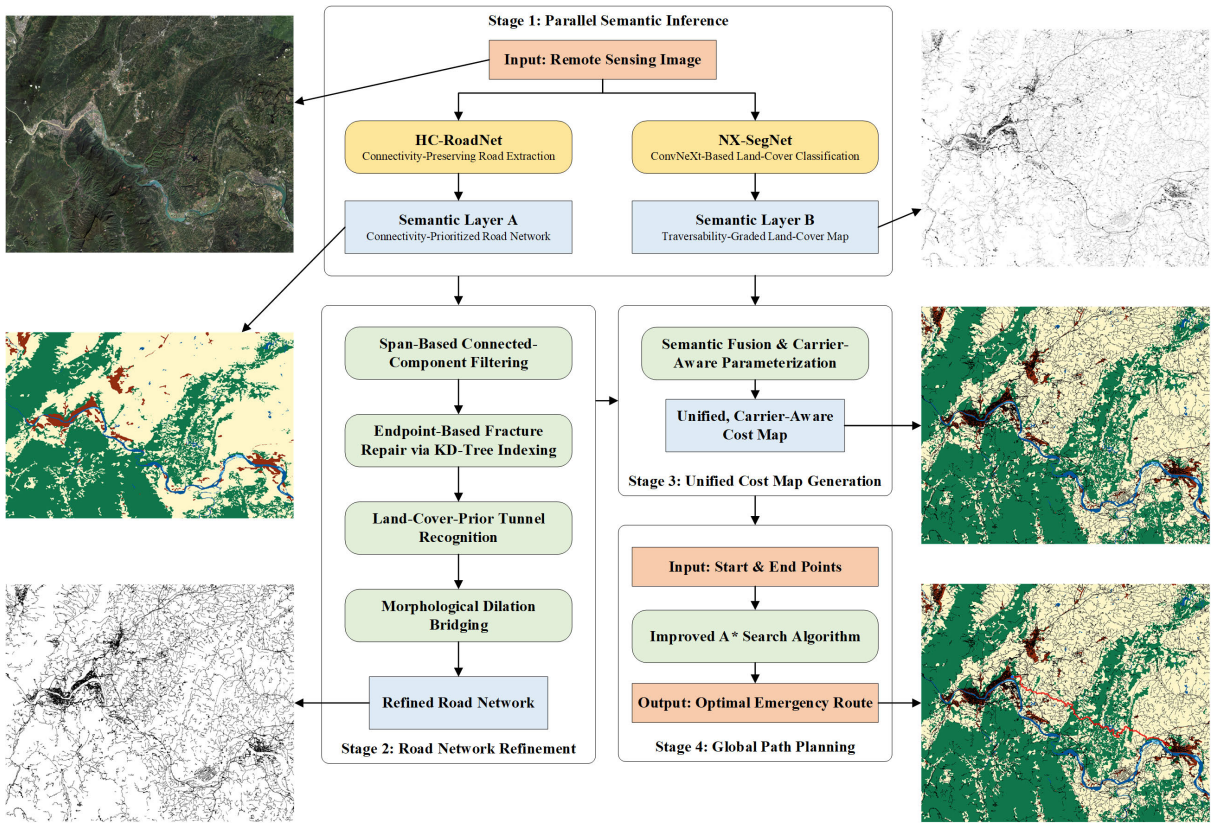

Fig. 4. Overall framework of the proposed carrier-aware global emergency routing methodology.

III. Methodology

A. Framework

The proposed methodology establishes an end-to-end framework for carrier-aware global emergency routing, designed to transform a single remote sensing image into an optimal and navigable path. The entire process, as illustrated in Fig. 4, is structured into four sequential stages: parallel semantic inference, road network refinement, unified cost map generation, and global path planning.

The core of our approach is to deconstruct the complex routing problem into distinct and manageable components. The

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pipeline initiates with a single, high-resolution remote sensing image of the area of interest, which serves as the sole input to the system. In Stage 1, this image is simultaneously processed by two specialized deep neural networks. The first stream employs HC-RoadNet (see Section III-B) to perform road network extraction, with a strong emphasis on preserving topological connectivity; the output is a binary mask representing a connectivity-prioritized road layer. The second stream concurrently utilizes NX-SegNet (see Section III-C) for fine-grained, multiclass land-cover classification, generating a traversability-graded land-cover layer, in which each class corresponds to a different level of off-road mobility. In Stage 2, the raw road mask undergoes a four-step refinement pipeline (see Section III-D) that systematically restores its topological connectivity. Span-based connected-component filtering removes isolated noise, endpoint-based fracture repair stitches small gaps, land-cover-prior tunnel recognition recovers occluded passageways across forest cover, and morphological dilation bridging closes residual subpixel breaks. This refinement step is essential because HC-RoadNet, by design, prioritizes high recall over high precision, leaving the raw output with a number of false positives and minor topological gaps that must be reconciled before downstream planning. In Stage 3, the refined road layer and the land-cover layer are fused into a single unified cost map (see Section III-E). This map assigns a specific traversal cost to each pixel under a road-priority logic, where roads serve as high-speed corridors that override the underlying land-cover impedance. The cost assignment is parameterized to be carrier-aware, allowing costs to be adjusted based on the physical characteristics of the emergency response platform. This fusion is critical, as it allows traversable terrain to act as a functional bridge across residual discontinuities in the extracted road network. Finally, in Stage 4, given a start–goal pair, an improved A* search algorithm augmented with a turn penalty and an optional road-priority hierarchical strategy (see Section III-F) computes the optimal route on the unified cost map.

B. HC-RoadNet: Connectivity-Preserving Road Extraction Network

The primary objective of the road extraction module is to generate a binary segmentation mask that not only achieves high pixelwise accuracy but, more importantly, preserves the topological connectivity of the road network. To this end, we developed HC-RoadNet, which encompasses a tailored network architecture, a connectivity-oriented loss function, and a robust training-and-inference protocol. As illustrated in Fig. 5, HC-RoadNet follows an encoder–decoder framework engineered to capture both multiscale contextual information and fine-grained spatial details crucial for distinguishing roads from complex backgrounds.

We employ EfficientNet-B7 [35], pretrained on the ImageNet dataset, as the feature encoder. Its deep and efficient architecture serves as a strong backbone, extracting a rich hierarchy of feature representations from the input image. We utilize a differential learning rate strategy, setting the encoder's learning rate lower than the decoder's, to leverage the pretrained knowledge while fine-tuning it for the road segmentation task.

The decoder is designed to progressively upsample the encoded features and integrate them with high-resolution features from the encoder via skip connections. To enhance this fusion process and improve feature representation, we have incorporated several key components.

1) Attention Gates (AGs): Positioned at each skip connection, AGs [36] automatically learn to focus on road-relevant structures. Given a feature map x^l from the decoder's layer l and a corresponding feature map g from a deeper layer, the attention coefficient α is computed as follows:

$$\alpha = \sigma_2\left(\psi^T\left(\sigma_1\left(W_x^T x^l + W_g^T g + b_g\right)\right) + b_\psi\right) \tag{1}$$

where W_x, W_g, and ψ are the linear transformations implemented as 1×1 convolutions, σ_1 is an SiLU activation function [37], and σ_2 is a Sigmoid function that scales the coefficients to a range of [0,1]. The output of the gate, $\hat{x}^l$, is the elementwise multiplication of the input feature map with the attention coefficients: $\hat{x}^l = x^l \otimes \alpha$. This process effectively suppresses irrelevant background features before they are fused with the decoder's feature maps.

2) Concurrent Spatial and Channel Squeeze-and-Excitation (scSE) Blocks: Integrated within the decoder's residual blocks, scSE modules [38] perform dynamic feature recalibration. For a given feature map $\mathbf{U} \in \mathbb{R}^{C\times H\times W}$, scSE computes a channelwise recalibration vector s_c and a spatialwise recalibration map s_s. The channel attention branch (cSE) uses global average pooling to produce channel statistics, while the spatial attention branch (sSE) uses a 1×1 convolution to generate a spatial saliency map. The final output $\hat{\mathbf{U}}$ is a combination of both

$$\hat{\mathbf{U}} = \mathbf{U} \otimes \mathbf{s}_c + \mathbf{U} \otimes \mathbf{s}_s. \tag{2}$$

This enhances informative features while diminishing less useful ones, which is particularly effective for highlighting thin, elongated road structures.

3) Residual Connections: Each upsampling stage in the decoder is followed by residual blocks [39], which facilitate gradient flow during training and enable the learning of more robust feature representations.

Standard segmentation losses are often insufficient to enforce topological correctness. To directly address this, we designed a composite loss function, ComboLoss, that explicitly optimizes for network connectivity. Our ComboLoss is a weighted sum of three components

$$L_{total} = \omega_1 L_{BCE} + \omega_2 L_{Tversky} + \omega_3 L_{clDice} \tag{3}$$

where $\omega_1 = 0.3, \omega_2 = 0.3$, and $\omega_3 = 0.4$ in our experiment. These weighting coefficients were determined through empirical trials to strike an optimal balance between pixel-level classification accuracy and structural continuity, with a slightly higher weight assigned to L_{clDice} to prioritize the topological integrity of the extracted road network.

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4) Weighted BCE Loss L_{BCE}: To address class imbalance, we use a weighted binary cross entropy. For a given prediction $\hat{y}$ and GT $y \in \{0, 1\}$, the loss is

$$L_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}\left[\omega_p \cdot y_i \log\left(\hat{y}_i\right) + (1 - y_i)\log\left(1 - \hat{y}_i\right)\right] \quad (4)$$

where $\omega_p = N_{neg}/N_{pos}$ is the positive weight, calculated as the ratio of nonroad to road pixels in the training set.

5) Tversky Loss $L_{Tversky}$: The Tversky index (TI) is defined as follows:

$$TI(\alpha, \beta) = \frac{\sum TP}{\sum TP + \alpha \sum FP + \beta \sum FN} \quad (5)$$

where TP, FP, and FN are the sums of true positive, false positive, and false negative over all pixels, respectively. By setting $\alpha + \beta = 1$ and choosing $\alpha < \beta$ (specifically, $\alpha = 0.3, \beta = 0.7$), we penalize FN more heavily than FP. The loss is then $L_{Tversky} = 1 - TI$.

6) Soft-clDice Loss L_{clDice}: This is the cornerstone of our connectivity-preserving strategy. This directly penalizes topological errors, such as gaps and breaks, encouraging the network to produce continuous road segments. It operates on soft-skeletonized versions of the prediction (Skel($\hat{Y}$)) and the GT (Skel(Y)). It is defined through two metrics: topological precision ($Tprec$) and topological sensitivity ($Tsens$)

$$Tprec = \frac{|\text{Skel}\left(\hat{Y}\right) \cap Y|}{|\text{Skel}\left(\hat{Y}\right)|} \quad \text{and} \quad Tsens = \frac{|\text{Skel}\left(Y\right) \cap \hat{Y}|}{|\text{Skel}\left(Y\right)|}. \quad (6)$$

The final clDice score is the harmonic mean of these two metrics, and the loss is

$$L_{clDice} = 1 - 2 \cdot \frac{Tprec \cdot Tsens}{Tprec + Tsens}. \quad (7)$$

Finally, we adopt a robust training and inference strategy. The model is trained using the AdamW optimizer [40] with a cosine annealing scheduler [41] with warm restarts to ensure stable convergence. An extensive online data augmentation pipeline—including random resized cropping, flips, random affine transformations (rotation, scaling, translation, and shear), and perspective shifts—is employed for generalization. We utilize automatic mixed precision [42] for faster training and maintain an exponential moving average [43] of the model's weights for more stable evaluation. During inference, test-time augmentation [44] is used to enhance the prediction robustness. The raw output of HC-RoadNet is then passed to the refinement pipeline described in Section III-D for downstream planning use.

C. NX-SegNet: ConvNeXt-Enhanced Land-Cover Classification Network

To effectively capture multiscale semantic features while preserving precise spatial localization for segmentation tasks, we propose a novel encoder–decoder architecture termed NX-SegNet. The detailed network structure is illustrated in Fig. 6. The network adopts an asymmetric U-shaped design, integrating a powerful convolutional backbone with a coordinate-aware attention mechanism to enhance the feature representation. The architecture consists of three principal components: a ConvNeXt-based encoder for hierarchical feature extraction, a CA bridge for global context embedding, and an attention-guided decoder for resolution recovery.

1) ConvNeXt-Based Encoder: The encoder utilizes ConvNeXt-Tiny [45] pretrained on ImageNet [46] as the backbone. ConvNeXt modernizes the standard ResNet architecture by incorporating design choices from Vision Transformers, such as larger kernel sizes and layer normalization, thereby offering a strong inductive bias for visual recognition. We extract feature maps from four distinct stages of the encoder, denoted by $\{E_1, E_2, E_3, E_4\}$, with channel dimensions corresponding to $\{96, 192, 384, 768\}$. These multilevel features provide a rich combination of low-level textural details and high-level semantic abstractions.

2) CA: To address the limitation of standard convolutional neural networks (CNNs) in capturing long-range dependencies and precise positional information, we introduce the CA module at the bottleneck and within each decoder block of NX-SegNet. Unlike global average pooling which compresses spatial information into a single scalar, the CA module [47] decomposes channel attention into two parallel 1-D feature encoding processes. For an input feature tensor $X \in \mathbb{R}^{C\times H\times W}$, the module performs pooling kernels $(H, 1)$ and $(1, W)$ to encode each channel along the vertical and horizontal coordinates, respectively. The output of the cth channel at height h and width w is formulated as follows:

$$z_h^c(h) = \frac{1}{W}\sum_{0\le i\le W} x_c(h, i), z_w^c(w) = \frac{1}{H}\sum_{0\le j\le H} x_c(j, w). \quad (8)$$

The resulting aggregated feature maps z_h and z_w are concatenated along the spatial dimension and transformed by a shared 1×1 convolution function F_1, followed by a nonlinear activation function δ (Hard-Swish) [48] to generate an intermediate feature map $f \in \mathbb{R}^{C/r\times(H+W)}$, where r is the reduction ratio. This intermediate tensor is then split along the spatial dimension into two separate tensors, f_h and f_w. These are subsequently transformed by two 1×1 convolutions F_h and F_w and a Sigmoid activation σ to produce the attention weights g_h and g_w. The final output Y is obtained by reweighting the input feature map using the broadcasted attention values

$$y_c(i, j) = x_c(i, j) \times \sigma\left(F_h\left(f_h^c(i)\right)\right) \times \sigma\left(F_w\left(f_w^c(j)\right)\right). \quad (9)$$

This mechanism allows the network to attend to large-scale regions of interest while retaining precise spatial awareness.

3) Enhanced Decoder Block: The decoder aims to reconstruct the segmentation map by progressively upsampling the deep features. We design an enhanced decoder block that fuses high-level semantics with low-level spatial details via skip connections. For the ith decoding stage, the feature map D_{i+1} from the previous layer is first upsampled via bilinear interpolation to match the resolution of the corresponding encoder feature map E_i. To resolve the channel mismatch between the encoder and decoder, a 1×1 convolution W_{skip} is applied to E_i. The aligned features are concatenated and processed by two consecutive 3×3 convolutional layers with batch normalization and rectified linear unit (ReLU) activation,

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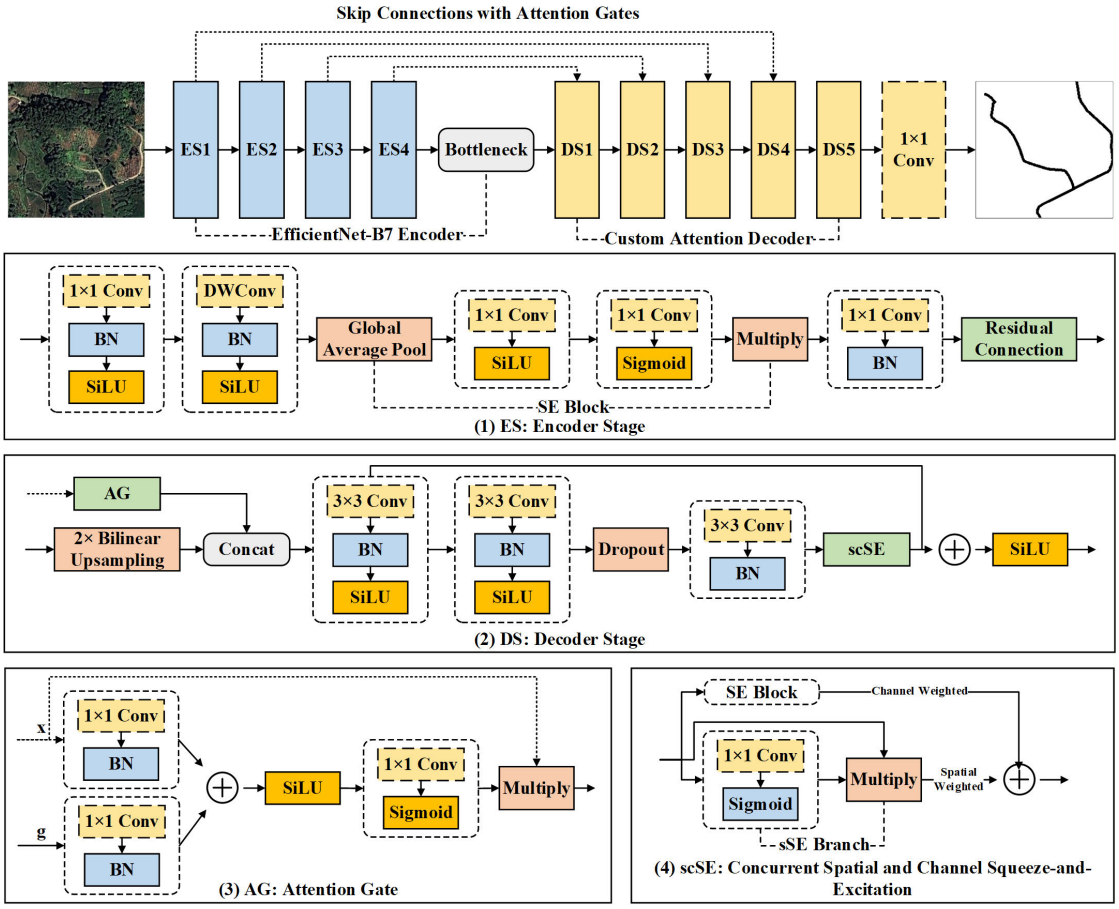

Fig. 5. Architecture of the HC-RoadNet.

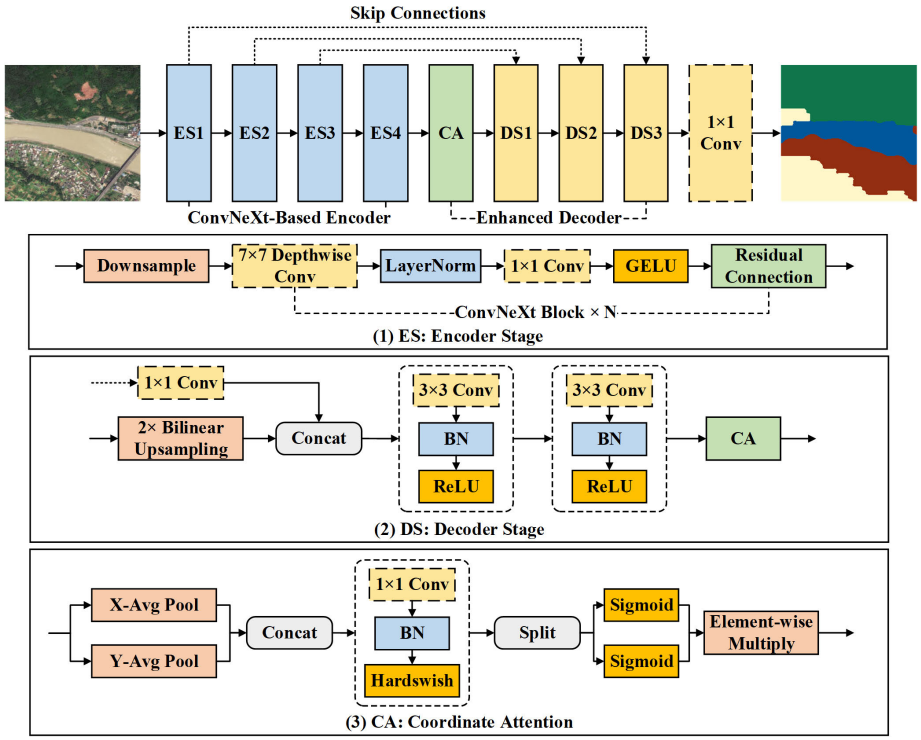

Fig. 6. Architecture of the NX-SegNet.

denoted by F_{conv}. Finally, the fused features are refined by the CA module to suppress background noise. The operation for a decoder block is defined as follows:

$$D_i = \mathrm{CA}\left(F_{conv}\left(\mathrm{Concat}\left(\mathrm{Upsample}\left(D_{i+1}\right), W_{skip}\left(E_i\right)\right)\right)\right). \quad (10)$$

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The final decoding stage is followed by a segmentation head consisting of convolutional layers to map the feature space to the N target classes. An interpolation operation is applied at the output to ensure the prediction matches the original input image resolution.

4) Loss Function and Optimization: To effectively supervise the learning process of NX-SegNet, we formulate the semantic segmentation task as a pixelwise multiclass classification problem. We employ the standard cross-entropy loss L_{CE} to quantify the divergence between the predicted probability maps and the GT annotations. Formally, for a given input image with M pixels and N target classes (where $N = 10$ in this study), the objective function is minimized over the training batch

$$L_{CE} = -\frac{1}{M}\sum_{i=1}^{M}\sum_{c=1}^{N} y_{i,c}\log\left(p_{i,c}\right) \tag{11}$$

where $y_{i,c} \in \{0, 1\}$ denotes the binary label indicating whether class c is the true category for pixel i, and $p_{i,c}$ represents the predicted probability derived from the Softmax activation function.

D. Road Network Refinement Pipeline

The raw output of HC-RoadNet, while exhibiting high recall and strong completeness (see Section IV-B), retains two characteristic imperfections that prevent its direct use in routing. The first comprises isolated false-positive blobs scattered across nonroad regions, originating from spectral confusion with road-like textures. The second comprises subpixel-to-several-pixel topological gaps along true road centerlines, caused by tree-canopy occlusion, cast shadows in mountainous terrain, and tunnel-induced visual occlusion. Both imperfections are detrimental to A*-based planning. False positives consume search time and can mislead the route through phantom corridors, while topological gaps are interpreted by the planner as hard barriers, forcing costly off-road detours.

To convert the raw mask $M_R \in \{0, 1\}^{H\times W}$ into a planning-ready road layer $M_{R'}$, we propose a four-step refinement pipeline that exploits geometric, spatial, and semantic priors. The pipeline operates entirely as a postprocessing module on the inferred mask and requires no additional training. We emphasize that, by performing this refinement at the cost-map level rather than within the network's loss function, we preserve the modularity of the parallel-inference paradigm described in Section III-A.

1) Span-Based Connected-Component Filtering: Direct area-based thresholding is suboptimal for road masks because true secondary roads, although having small total pixel area, exhibit elongated spatial extent. We therefore filter connected components by their spatial span rather than their area. Let $C = \{c_1, c_2, \ldots, c_n\}$ be the set of eight-connected components of M_R. For each component c_i, let W_i and H_i be the width and height of its minimum bounding rectangle. A component is retained if and only if it spans at least τ_s pixels in either the horizontal or vertical direction

$$f_{\text{filter}}(c_i) = \begin{cases} c_i & \text{if } (W_i \geq \tau_s) \vee (H_i \geq \tau_s) \\ \emptyset & \text{otherwise.} \end{cases} \tag{12}$$

The threshold τ_s is set to 20% of the input tile size, i.e., $\tau_s = \lfloor 256 \times 0.2 \rfloor = 52$ pixels. This value was determined empirically by sweeping τ from 10 to 100 and selecting the value maximizing intersection over union (IoU) on a validation subset; the resulting curve is reported in Section IV-D.

2) Endpoint-Based Fracture Repair via KD-Tree Indexing: Small topological gaps are repaired by detecting unmatched skeleton endpoints and connecting them under strict geometric and topological constraints. The procedure consists of three phases.

Skeleton Extraction: We apply the Zhang-Suen thinning algorithm to obtain a single-pixel-width skeleton S of the filtered mask. This C++-optimized implementation reduces skeletonization time by approximately two orders of magnitude compared with iterative morphological erosion in pure Python.

Endpoint Detection by Local Convolution: An endpoint is defined as a foreground skeleton pixel with exactly one foreground neighbor in its 3×3 neighborhood. Detection is performed by convolving S with the kernel

$$K = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 10 & 1 \\ 1 & 1 & 1 \end{bmatrix}. \tag{13}$$

A pixel p is identified as an endpoint if and only if $(S^{*}K)(p) = 11$. Let E be the resulting endpoint set.

Constrained Pairing via KD-Tree: We construct a KD-Tree spatial index over E to retrieve all endpoint pairs (e_i, e_j) within Euclidean distance $D_{th} = 20$ pixels in $O(N \log N)$ time, in contrast to the $O(N^2)$ of brute-force enumeration. A candidate pair is accepted as a fracture-repair edge only if both of the following constraints are satisfied.

1) *Uniqueness:* e_i has exactly one neighbor within radius D_{th}, preventing ambiguous multiway junctions from being collapsed.
2) *Cross-Component Requirement:* e_i and e_j belong to different connected components in the original mask. This constraint, enforced via a precomputed component label matrix L, prevents self-loops at small skeleton burrs within the same component.

For each accepted pair, a 2-pixel-wide line segment is drawn between e_i and e_j, completing the local closure. The full-algorithmic procedure is summarized in Algorithm 1.

The choice $D_{th} = 20$ pixels balances repair coverage against the risk of erroneous bridging. As reported in Section IV-D, increasing D_{th} to 30 or 40 marginally raises completeness but degrades correctness, indicating spurious cross-component connections; we therefore retain $D_{th} = 20$ as the operating point. Because emergency planning prizes topological correctness over completeness, since a phantom shortcut may divert rescue assets into impassable terrain, we explicitly accept the tighter D_{th}.

3) Land-Cover-Prior Tunnel Recognition: A subset of road fractures is induced not by image noise but by genuine physical occlusion. Tunnels passing through forested mountain ridges produce extended gaps that nonetheless correspond to real and traversable infrastructure. To recover these, we exploit the land-cover prior produced by NX-SegNet,

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Algorithm 1 Endpoint-Based Fracture Repair

Input: binary road mask M_R
distance threshold D_{th}
Steps: $L \leftarrow$ ConnectedComponents(M_R)
$S \leftarrow$ ZhangSuenThinning(M_R)
$E \leftarrow \{p \in S \mid (S^*K)(p) = 11\}$
$T \leftarrow$ BuildKDTree(E)
$P \leftarrow T$.QueryPairs(D_{th})
For each endpoint e_i, build its neighbor list $N[e_i]$ from P
$C \leftarrow \emptyset$// accepted fracture edges
for each $e_i \in E$ **do**
if $|N[e_i]| = 1$ **then**
$e_j \leftarrow N[e_i][0]$
if $L(e_i) \neq L(e_j)$ **then** $C \leftarrow C \cup \{(e_i, e_j)\}$
end if
end for
$M_R^{(2)} \leftarrow$ Copy(M_R)
for each $(e_i, e_j) \in C$ **do**
DrawLine($M_R^{(2)}, e_i, e_j$, width = 2)
end for
return $M_R^{(2)}$
Output: fracture-repaired road mask $M_R^{(2)}$

leveraging the empirical observation that tunnels are predominantly constructed on high-grade trunk roads passing under forest-covered terrain. The procedure consists of three phases:

Trunk-Road Isolation via Asymmetric Morphology: We first dilate $M_R^{(2)}$ with a small structuring element K_d (3×3) to absorb minor noise; this dilated state defines the reference connectivity C_{ref}. We then apply an aggressive erosion with a larger structuring element K_e (11×11) to suppress secondary lanes and retain only trunk-road backbones M_{main}

$$M_{\text{dilated}} = M_R^{(2)} \oplus K_d \tag{14}$$

$$M_{\text{main}} = M_{\text{dilated}} \ominus K_e. \tag{15}$$

A skeleton I_{skel} and endpoint set E_{main} are then extracted from M_{main} as in step 2.

Forest-Mask Spatial Constraint: From the NX-SegNet output M_L, we extract a forest mask M_{forest} (Class 1). For each candidate endpoint pair (e_i, e_j) within distance D_{max} retrieved by KD-Tree, we apply three filters: 1) cross-component test: $L(e_i) \neq L(e_j)$ in C_{ref}; 2) noninternal test: at least one of e_i, e_j must lie outside M_{forest}; and 3) forest-overlap test: the line segment connecting e_i and e_j must overlap with M_{forest} by at least 20% of its length.

Best-Pair Selection per Component Pair: Where multiple candidate edges connect the same pair of components, only the shortest distance edge is retained

$$\left(e_i^*, e_j^*\right) = \arg\min_{e_i \in C_A, e_j \in C_B} \|e_i - e_j\|_2. \tag{16}$$

This produces the final tunnel-edge set T, which is rasterized into the mask to yield $M_R^{(3)}$.

4) Morphological Dilation Bridging: Despite steps 1–3, residual single- to two-pixel-wide breaks may persist due to imaging artifacts or tile-boundary effects. Because the planner treats any single-pixel break as a hard barrier, we apply a final morphological dilation with a circular structuring element B of diameter $d = 5$ pixels

$$M_{R'}(p) = \max_{q \in N_B(p)} M_R^{(3)}(q). \tag{17}$$

This isotropic expansion bridges subpixel breaks at the cost of slight perceptual broadening of road segments. We emphasize that this broadening has no adverse effect on routing. In the cost-map formulation (see Section III-E), broadening the road class merely enlarges the set of low-cost pixels available to the planner, which does not change the topology or homotopy class of optimal paths.

The output $M_{R'}$ is the final, planning-ready road layer. The quantitative ablation of each step is reported in Section IV-D. The experimental results represent a substantial recovery of precision-side metrics that directly addresses the recall–precision tradeoff intrinsic to HC-RoadNet's connectivity-prioritized design.

E. Carrier-Aware Parameterization and Unified Cost Map Generation

Building on the refined road layer $M_{R'}$ from Section III-D and the land-cover map M_L from NX-SegNet, we construct a unified cost map that integrates the topological advantages of road networks with the traversability of land-cover features. Let $I \in \mathbb{R}^{H \times W \times 3}$ be the input remote sensing image. The preceding modules generate two semantic outputs: the binary road extraction mask $M_R \in \{0, 1\}^{H \times W}$ and the multiclass land-cover classification map $M_L \in \{0, \ldots, 9\}^{H \times W}$.

1) Unified Semantic Fusion: The fused semantic map $S_{unified}$ is derived by prioritizing the road network layer over the land-cover layer. Since roads serve as high-speed corridors, any pixel identified as a road in M'_R overwrites the underlying land-cover classification

$$S_{unified}(p) = \begin{cases} 10, & \text{if } M'_R(p) = 1 \\ M_L(p), & \text{otherwise} \end{cases} \tag{18}$$

where the index 10 is reserved to represent the "highway" class.

2) Carrier-Specific Cost Parameterization: Once the unified semantic map is obtained, it must be transformed into weights interpretable by the path-search algorithm. Conventional path planning typically assumes terrain isotropy; however, in emergency rescue scenarios, different rescue carriers exhibit drastically different sensitivities to the impedance of the same land-cover type. To this end, we propose a carrier-aware traversal cost mapping model that converts semantic attributes into a generalized traversal cost. We define the traversal cost at pixel p of $S_{unified}$ for a specific carrier k as $G(p, k)$. The cost function is composed of a weighted combination of travel time and energy consumption, modeling the dual demands of efficiency and endurance in rescue missions

$$G(p, k) = \alpha \cdot \frac{1}{V(T(p), k)} + \beta \cdot E(T(p), k) \tag{19}$$

where $V(T(p), k)$ denotes the expected average movement speed of carrier k on land-cover type $T(p)$, $E(T(p), k)$ denotes the corresponding per-unit-distance energy consumption coefficient, and α, β are weighting coefficients that balance the

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mission's emphasis on speed versus energy. For different rescue carriers, distinct parameterization schemes can be designed according to real-world conditions, thereby transforming the cost map from a static attribute layer into a dynamic weight matrix tightly coupled with the mission carrier. By convention, when $E = \infty$ the pixel is impassable under any optimization mode that includes a nonzero β. Under pure time-priority ($\beta = 0$), the pixel becomes impassable only if $V = 0$ as well; otherwise it remains traversable at a cost of α/V. This dual encoding allows the table entries to distinguish between absolute physical constraints ($V = 0$) and soft constraints that may be relaxed under extreme time pressure ($V > 0$). Concrete instances are elaborated in Section IV-E1.

F. Improved a With Turn Penalty and Road-Priority Hierarchical Strategy*

With the unified cost map $G(p, k)$ established, we formulate the emergency routing problem as finding the minimum-cumulative-cost path $P^* = \{p_1, p_2, \ldots, p_n\}$ on a grid-based graph from a start node p_s to a goal node p_g. We employ the A* algorithm as the backbone search procedure, augmented with two task-specific enhancements.

1) Graph Representation and Cost-Aware Heuristic: The search space is modeled as an eight-connected grid. The A* evaluation function is augmented as follows:

$$f(p) = g(p) + \varepsilon \cdot h(p) + \theta\left(\Delta\varphi_p\right). \tag{20}$$

In this formulation, $g(p)$ is the accumulated traversal cost from p_s to p, integrated along the path with edge weights w_d (1.0 for cardinal moves, $(2)^{1/2} \approx 1.414$ for diagonal moves), calculated as follows:

$$g(p_n) = \sum_{i=0}^{n-1} w_d\left(p_i, p_{i+1}\right) \cdot \frac{G(p_i, k) + G\left(p_{i+1}, k\right)}{2}. \tag{21}$$

Furthermore, $h(p)$ represents the Euclidean distance heuristic to p_g, while ε is a per-carrier admissibility constant defined as the minimum unit cost across all reachable terrain classes [49]. Multiplying $h(p)$ by ε guarantees that the heuristic never overestimates the true minimum-cost-to-go, preserving A*'s optimality. Finally, $\theta(\Delta\varphi_p)$ serves as a turn penalty, where $\Delta\varphi_p$ denotes the angle between the incoming direction and the outgoing direction. This specific component encourages physically smooth trajectories by suppressing saw-tooth artifacts.

The full search procedure is summarized in Algorithm 2.

2) Road-Priority Hierarchical Strategy for High-Mobility Carriers: For carriers strongly biased toward roads, we adopt a three-stage hierarchical decomposition.

Ingress (Start → Road): A local A* search from p_s to the nearest access point $p_a \in M_{R'}$ (found via KD-Tree).

Transit (Road → Road): Search operates on the road subgraph from p_a to an exit point p_e near the goal.

Egress (Road → Goal): A final local A* search connects p_e to p_g.

The full path is the concatenation

$$P^* = A^*(p_s, p_a) \oplus A^*(p_a, p_e) \oplus A^*\left(p_e, p_g\right). \tag{22}$$

Algorithm 2 Improved A* With Turn Penalty

Input: cost map G; start node p_s; goal node p_g; admissibility constant ε; turn penalty function θ

Steps: OpenSet ← priority queue
push $(f = \varepsilon \cdot h(p_s), p_s, d_0 = 0)$ into OpenSet
$g(p_s) \leftarrow 0$; $g(\text{others}) \leftarrow \infty$; Parents ← $\emptyset$
while OpenSet not empty **do**
$(f, p_i, d_{i-1}) \leftarrow$ OpenSet.PopMin()
if $p_i = p_{\text{goal}}$ **then return**
ReconstructPath(Parents, p_g)
for each $p_{i+1} \in$ Neighbors(p_i) **do**
$d_i \leftarrow p_{i+1} - p_i$
$g_{\text{move}} \leftarrow w_d(p_i, p_{i+1}) \cdot [G(p_i) + G(p_{i+1})]/2$
$g_{\text{turn}} \leftarrow \theta(\text{angle}(d_{i-1}, d_i))$
$g_{\text{tent}} \leftarrow g(p_i) + g_{\text{move}} + g_{\text{turn}}$
if $g_{\text{tent}} < g(p_{i+1})$ **then**
Parents$(p_{i+1}) \leftarrow p_i$
$g(p_{i+1}) \leftarrow g_{\text{tent}}$
$h \leftarrow \varepsilon \cdot \|p_{i+1} - p_{\text{goal}}\|_2$
Push $(g_{\text{tent}} + h, p_{i+1}, d_i)$ into OpenSet
end if
end for
end while
return failure

Output: optimal path from p_s to p_g, or failure

This decomposition reduces effective complexity because the most expensive segment, the long-range cross-region transit, is solved on the sparse road subgraph rather than the full grid. KD-Tree queries provide $O(\log n)$ access-point lookup. For carriers with low road dependence, the hierarchical strategy is bypassed. If the road network is disconnected, the system falls back to direct full-grid A*.

IV. Experimental Results and Analysis

A. Implementation Details

The computational infrastructure utilized in this study consisted of an Intel Core i7-14700K processor operating at 3.40 GHz, paired with 64-GB RAM and an Nvidia GeForce RTX 5090 GPU (32 GB). The software implementation was built upon Python 3.12.11 and the PyTorch 2.8.0+cu129 framework. Auxiliary libraries employed included NumPy 2.1.2, SciPy 1.16.2, Scikit-learn 1.7.2, Matplotlib 3.10.6, and Seaborn 0.13.2 to ensure reproducibility of the results. The KD-Tree spatial indexing used in the road refinement pipeline (see Section III-D) and the road-priority hierarchical planner (see Section III-F) was implemented via scipy.spatial.cKDTree.

Both HC-RoadNet and NX-SegNet were trained from ImageNet-pretrained backbones using the AdamW optimizer with weight decay 1×10^{-2}, initial learning rate 1×10^{-4}, and a cosine annealing schedule with warm restarts. HC-RoadNet was trained for 50 epochs on STRoadSet; NX-SegNet was trained for 100 epochs on STLandSet. An early stopping criterion (patience = 15 epochs on validation loss) was applied to both models. To eliminate implementation bias in the

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Fig. 7. Road network extraction results of HC-RoadNet in the target study area.

TABLE I
QUANTITATIVE COMPARISON OF ROAD EXTRACTION RESULTS

Model	IoU	Precision	Recall	F1	Correctness	Completeness	Relaxed_F1
UNet	0.253	0.6358	0.2959	0.4039	0.8532	0.4851	0.6185
UNet++	0.2476	0.6447	0.2868	0.397	0.8636	0.4715	0.61
MA-net	0.2317	0.6481	0.2651	0.3762	0.8584	0.4525	0.5926
Linknet	0.2464	0.6498	0.2842	0.3954	0.8697	0.498	**0.6333**
FPN	0.2111	**0.6839**	0.2339	0.3485	**0.904**	0.4257	0.5788
PSPNet	0.2502	0.5983	0.3007	0.4002	0.8467	0.4909	0.6215
PAN	0.1891	0.6448	0.2111	0.318	0.8738	0.3737	0.5235
DeepLabV3+	0.2284	0.652	0.2602	0.3719	0.8771	0.4598	0.6033
HC-RoadNet	**0.2588**	0.2771	**0.7973**	**0.4112**	0.4865	**0.8807**	0.6267

comparison study, all baseline models (U-Net, U-Net++, MA-Net, LinkNet, FPN, PSPNet, PAN, and DeepLabV3 +) were reimplemented and retrained under an identical training protocol using the open-source segmentation_models.pytorch library, ensuring that performance differences arise from architectural and loss-function design rather than from hyperparameter or implementation discrepancies.

B. Performance on Road Extraction

This section evaluates HC-RoadNet on the STRoadSet testing set, both quantitatively against eight baseline architectures and through a component-level ablation study. The qualitative output of HC-RoadNet over the full study area is shown in Fig. 7, which closely aligns with the GT network in Fig. 1(b).

1) Evaluation Metrics: We adopt two complementary metric families. The strict pixel-level metrics (IoU, F1, precision, and recall) require pixel-perfect overlap and harshly penalize even subpixel boundary deviations, which is misaligned with the operational requirement that a road be merely traversable, not pixel-precise. We therefore additionally report buffer-based relaxed metrics: correctness (relaxed precision), completeness (relaxed recall), and the harmonic-mean Relaxed_F1, where the buffer is generated by morphological dilation with radius $r = 3$ pixels. Let P be the set of predicted road pixels, G be the set of GT road pixels, and $\delta(\cdot)$ be the morphological dilation operator. The relaxed metrics are

$$\text{Correctness} = \frac{|P \cap \delta(G)|}{|P|} \tag{23}$$

$$\text{Completeness} = \frac{|G \cap \delta(P)|}{|G|} \tag{24}$$

$$\text{Relaxed_F1} = 2 \cdot \frac{\text{Correctness} \cdot \text{Completeness}}{\text{Correctness+Completeness}}. \tag{25}$$

2) Quantitative Comparison: Table I reports the comparison between HC-RoadNet and eight mainstream segmentation networks. HC-RoadNet attains the highest strict IoU (0.2588), the highest strict F1 (0.4112), the highest recall (0.7973), and the highest completeness (0.8807). The Relaxed_F1 of 0.6267 is competitive with the best baseline (LinkNet, 0.6333) and

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TABLE II
ABLATION STUDY OF HC-ROADNET COMPONENTS ON STROADSET

ID	Configuration	AG	scSE	clDice	IoU	F1	Relaxed_F1
1	Baseline	✗	✗	✗	0.2369	0.3831	0.6186
2	+ AG + scSE	✓	✓	✗	0.2468	0.3959	0.6347
3	+ clDice only	✗	✗	✓	0.2320	0.3766	0.6118
4	Full HC-RoadNet	✓	✓	✓	0.2588	0.4112	0.6267

notably surpasses widely used architectures, such as U-Net (0.6185), PSPNet (0.6215), and DeepLabV3 + (0.6033).

3) On the Precision–Recall Asymmetry: The most striking pattern in Table I is the inversion of the precision–recall ordering between HC-RoadNet and the baselines. Baselines, such as FPN and DeepLabV3 +, exhibit high precision (0.65–0.68) but low recall (0.21–0.30), whereas HC-RoadNet shows the opposite asymmetry: precision 0.2771 against recall 0.7973. This pattern has previously been raised as a potential concern, and we address it directly here.

The asymmetry is a deliberate, designed property rather than a calibration artifact, and arises from the joint action of three loss components: 1) the positive-class weight ω_p in the BCE term (4), which up-weights the contribution of road pixels in proportion to the foreground/background imbalance; 2) the asymmetric Tversky configuration $\alpha = 0.3$ and $\beta = 0.7$ (4), which penalizes false negatives roughly 2.3× more than false positives; and 3) the soft-clDice term (7), which optimizes for skeletal continuity rather than boundary sharpness. Together these terms steer the network toward a high-recall, wide-boundary regime: the model recovers the vast majority of the topological structure but does so with slightly thicker road masks than the GT.

This asymmetry is operationally justified for emergency response. The cost of a false negative, namely, failing to recover a road segment that in fact exists, is the potential isolation of a population in need of rescue. The cost of a false positive, namely, predicting a road pixel where none exists, is in principle more recoverable. The spurious pixels manifest either as boundary thickening adjacent to true roads (which the planner ignores due to homotopy invariance) or as isolated false-positive blobs (which the refinement pipeline of Section III-D removes by span-based filtering). The spatial distribution of the false positives also matters. Empirically, we observe that the false-positive pixels of HC-RoadNet are concentrated near true road structures (as boundary thickening) and at scattered isolated locations (susceptible to span-based filtering), rather than being randomly distributed across the image. This distribution is precisely what makes the refinement pipeline effective at reclaiming precision without sacrificing recall, as we will demonstrate in Section IV-D.

4) Component-Level Ablation: To verify the contribution of each architectural and loss component of HC-RoadNet, we conducted an ablation study on STRoadSet. Four configurations were compared: a baseline EfficientNet-B7 U-shaped network without the proposed attention modules and without clDice; the same with AGs and scSE blocks but without clDice; the baseline with clDice but without attention; and the full HC-RoadNet. Results are reported in Table II.

Three observations follow from Table II. First, comparing rows 1 and 2, the addition of AG and scSE alone improves IoU by 0.99 percentage points (pp) and Relaxed_F1 by 1.61 pp, confirming that attention-based feature recalibration suppresses background distractors and sharpens road feature representation. Second, comparing rows 1 and 3, adding the clDice term in the absence of strong attention-based feature support actually degrades performance slightly (IoU 0.2369 → 0.2320). This indicates that the strict topological constraint imposed by clDice can conflict with pixelwise losses when the underlying feature representation is not sufficiently discriminative, since the network oscillates between satisfying the skeletal-continuity term and the pixel-level term. Third, comparing rows 2 and 4, when clDice is combined with attention modules, it produces a substantial IoU and F1 boost (+1.20 pp and +1.53 pp, respectively), with a small Relaxed_F1 reduction (−0.80 pp). The reduction reflects the intended effect of clDice. It tightens predictions toward the true skeleton, trimming the redundant boundary pixels that artificially inflate the buffer-based Relaxed_F1, while improving stricter metrics. The full HC-RoadNet thus achieves the best strict IoU and F1, and (crucially for the downstream planner) the most topologically faithful skeleton.

C. Performance on Land-Cover Classification

This section evaluates NX-SegNet on STLandSet with the same suite of baselines used for road extraction. The qualitative output is shown in Fig. 8.

1) Evaluation Metrics: Land-cover classification is a multiclass problem; we use mIoU as the primary indicator and additionally report macro-averaged accuracy, precision, recall, and F1 to ensure all classes contribute equally regardless of their pixel frequency

$$\text{mIoU} = \frac{1}{k}\sum_{i=1}^{k}\frac{p_{ii}}{\sum_{j=1}^{k} p_{ij} + \sum_{j=1}^{k} p_{ji} - p_{ii}} \tag{26}$$

where k is the number of classes and p_{ij} denotes the number of pixels of class i predicted as class j.

2) Quantitative Comparison: Table III reports the comparison.

NX-SegNet outperforms all baselines on every metric. The mIoU of 0.4875 surpasses the second-best (PAN, 0.4744) by 1.31 pp and the classical U-Net baseline by 2.51 pp. While these absolute gaps may appear modest, the consistent improvement across all five metrics indicates that NX-SegNet

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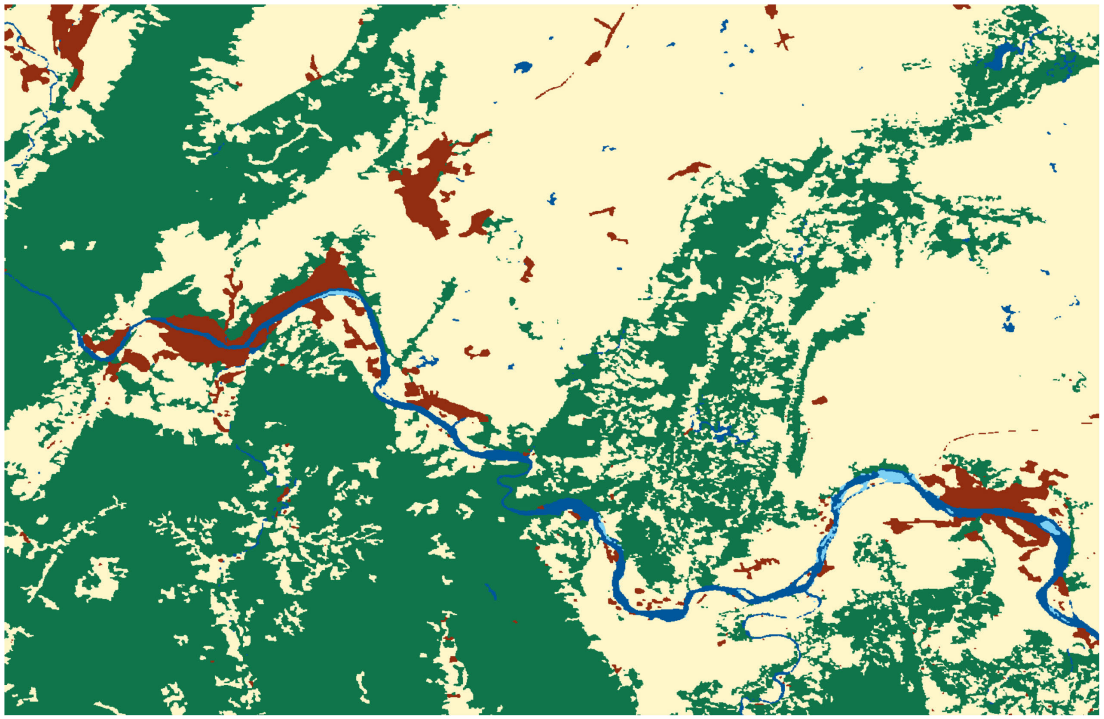

Fig. 8. Land-cover classification results of NX-SegNet in the target study area.

TABLE III
QUANTITATIVE COMPARISON OF LAND-COVER CLASSIFICATION RESULTS

Model	Accuracy	Precision	Recall	F1	mIoU
UNet	0.9895	0.4776	0.4825	0.4798	0.4624
UNet++	0.9923	0.4796	0.4895	0.4844	0.4705
MA-net	0.9804	0.4674	0.4289	0.4386	0.4061
Linknet	0.9904	0.4814	0.4506	0.4622	0.438
FPN	0.99	0.4861	0.4848	0.4854	0.4721
PSPNet	0.9834	0.4691	0.4795	0.4742	0.4524
PAN	0.9908	0.4843	0.4891	0.4867	0.4744
DeepLabV3+	0.9901	0.4834	0.4877	0.4855	0.4723
NX-SegNet	**0.9974**	**0.4932**	**0.4940**	**0.4936**	**0.4875**

TABLE IV
ABLATION STUDY OF NX-SEGNET COMPONENTS ON STLANDSET

ID	Configuration	Backbone	CA	Accuracy	mIoU
1	Baseline	ResNet-50	✗	0.9772	0.4482
2	+ ConvNeXt only	ConvNeXt-Tiny	✗	0.9923	0.4773
3	+ CA only	ResNet-50	✓	0.9708	0.3729
4	**Full NX-SegNet**	**ConvNeXt-Tiny**	✓	**0.9974**	**0.4875**

attains a balanced rather than skewed discriminative regime, in contrast with the deliberately asymmetric operating point of HC-RoadNet. Specifically, the gap between precision (0.4932) and recall (0.4940) is only 0.0008, indicating the model neither over-segments nor under-segments any class. This balanced behavior is desirable for land-cover classification because the downstream cost-map fusion (see Section III-E) treats each nonroad class as a candidate impedance level, and bias toward any single class would distort the spatial distribution of off-road traversal cost.

3) Component-Level Ablation: To attribute the improvement of NX-SegNet to its specific design choices, namely, the ConvNeXt-Tiny encoder and the CA bridge, we conducted

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TABLE V
STAGEWISE ABLATION OF THE ROAD NETWORK REFINEMENT PIPELINE

Stage	IoU	Precision	Recall	F1	Correctness	Completeness	Relaxed_F1
HC-RoadNet	0.2588	0.2771	0.7973	0.4112	0.4865	0.8807	0.6267
+ Span-based filtering	0.2634	0.2921	0.7281	0.4169	0.7578	0.8347	0.7944
+ Endpoint fracture repair	0.2631	0.2910	0.7326	0.4165	0.7578	0.8369	0.7954

an ablation study on STLandSet. Four configurations were compared. Results are reported in Table IV.

Two notable patterns emerge. Comparing rows 1 and 2, replacing ResNet-50 with ConvNeXt-Tiny while keeping the decoder fixed raises mIoU from 0.4482 to 0.4773 (+6.5%), confirming that the larger effective receptive field of the ConvNeXt 7×7 depthwise convolutions captures the coherent spatial extent of land-cover patches more effectively than the standard ResNet receptive field. Comparing rows 1 and 3, adding CA on top of the ResNet-50 backbone unexpectedly degrades mIoU (0.4482 $\rightarrow$ 0.3729). We attribute this regression to a feature-distribution mismatch between the coordinate-decomposition mechanism of CA and the locally biased features of ResNet-50: CA is designed to leverage the long-range spatial context provided by globally aware backbones, and when paired with a backbone whose features are local, the spatial attention maps it produces are noisy and dilute the discriminative signal. Comparing rows 2 and 4, when CA is paired with the ConvNeXt encoder, it provides a clear additional gain (mIoU 0.4773 $\rightarrow$ 0.4875, +1.0 pp), confirming the synergy between the global-receptive-field encoder and the coordinate-aware attention bridge. This synergy is a designed property of NX-SegNet rather than a fortuitous combination.

TABLE VI
EFFECT OF FRACTURE-REPAIR DISTANCE THRESHOLD

D_{th}	IoU	Correctness	Completeness	Relaxed_F1
20	0.2631	0.7578	0.8369	0.7954
30	0.2624	0.7560	0.8381	0.7950
40	0.2617	0.7543	0.8388	0.7943

TABLE VII
EFFICIENCY COMPARISON OF FRACTURE-REPAIR IMPLEMENTATIONS

Method	Skeletonization	Search Strategy	Runtime	Speedup
Conventional	morphological iteration	brute force	964.0 s	1.0×
Proposed	Zhang-Suen	KD-Tree	4.7 s	250.1×

Runtime is dominated by Zhang-Suen thinning and KD-Tree construction, both of which are independent of D_{th}. Varying $D_{th} \in \{20,30,40\}$ produced no measurable difference at the reported precision (worst-case 4.7 s in all three settings).

D. Effect of Road Network Refinement Pipeline

The refinement pipeline introduced in Section III-D is the primary mechanism by which the high-recall, wide-boundary output of HC-RoadNet is converted into a planning-ready road layer. This section quantifies its effect through a stagewise ablation and reports two implementation aspects (threshold sensitivity and runtime efficiency) that influence deployment.

1) Stagewise Ablation: Table V reports the effect of the first three refinement stages on the strict and relaxed metrics, evaluated on the same test region as Section IV-B. The fourth stage, morphological bridging, is applied as the final step to ensure pixel-level connectivity for the planner. Because isotropic dilation slightly broadens road masks, the buffer-based relaxed metrics would be biased by such broadening; we therefore exclude this stage from the metric-based ablation and discuss it qualitatively below.

Two observations stand out. First, span-based filtering accounts for the dominant share of the improvement. Correctness leaps from 0.4865 to 0.7578 (+27.13 pp) and Relaxed_F1 from 0.6267 to 0.7944 (+16.77 pp). This single step removes the false-positive blobs that were the principal contributor to HC-RoadNet's low Precision, validating the spatial-distribution argument made in Section IV-B that the false positives are predominantly small isolated structures rather than diffuse contamination. Second, the slight recall reduction at the filtering step (0.7973 $\rightarrow$ 0.7281) is expected, since the filter removes both true positives that happen to fall below the 52-pixel span threshold (typically very short, isolated road stubs disconnected from the main network) and false positives. Disconnected short stubs cannot anchor any feasible route in the planner, so their loss has no operational cost.

Endpoint fracture repair adds a small further gain in recall (0.7281 $\rightarrow$ 0.7326) and completeness (0.8347 $\rightarrow$ 0.8369) without harming Correctness, indicating that the cross component and uniqueness constraints (see Algorithm 1) effectively suppress spurious bridging. Tunnel recognition recovers extended occluded segments that the buffer-based metrics (with $r = 3$ pixels) cannot score directly but which are critical to network reachability; we illustrate two such cases in Fig. 9. Morphological bridging, applied last, contributes the final pixel-level closure required by the planner.

2) Implementation Notes: Two implementation choices warrant brief mention because they affect deployment but are not the central methodological claim. The span threshold τ_s was empirically swept across $[10, 100]$ pixels and yields a flat optimum near $\tau_s = 52$, consistent with the 20% tile-size design rationale. The endpoint-pairing distance threshold D_{th},

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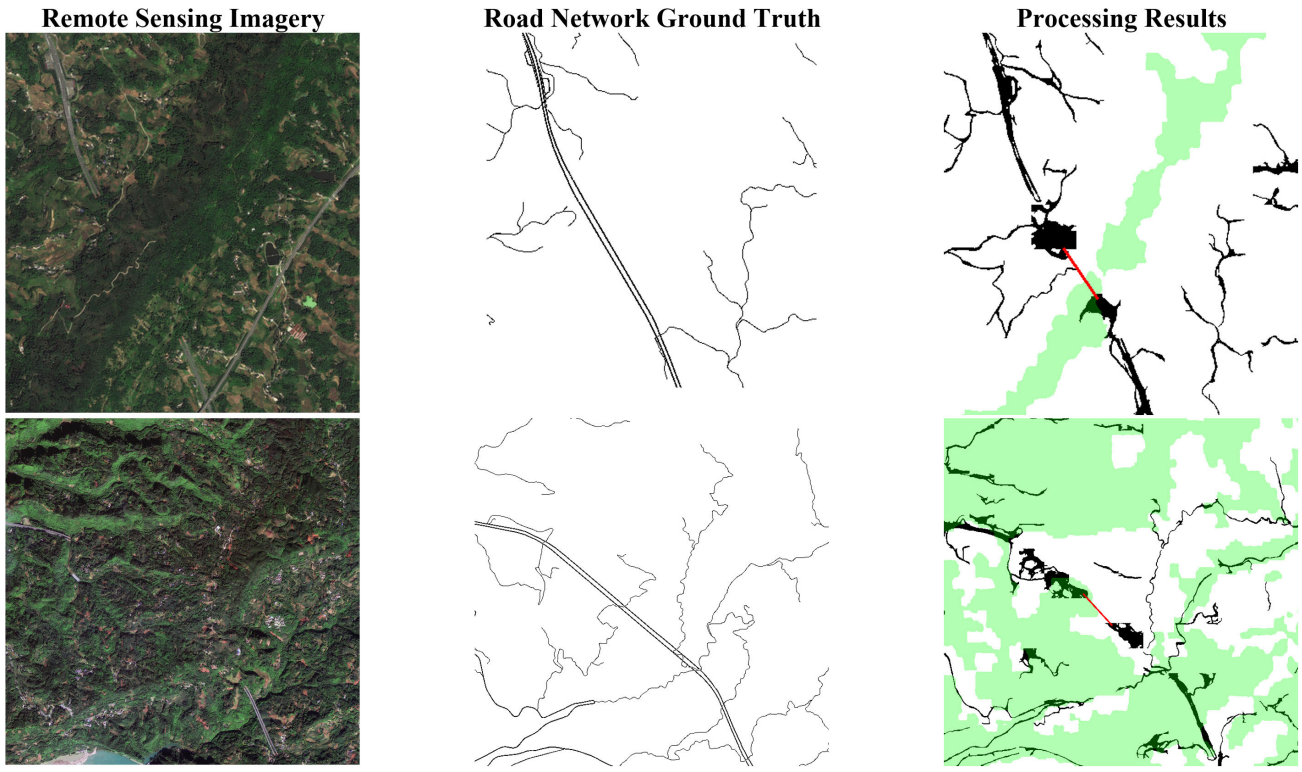

Fig. 9. Illustrative cases of tunnel recovery via land-cover prior. The left column shows the original satellite imagery with the tunnel occluded by forest canopy. The center column shows the GT road label, which depicts the actual through passage. The right column shows the refined mask with the recovered tunnel edge (red) overlaid on the forest mask (semitransparent green).

varied in $\{20, 30, 40\}$, shows a small but monotone tradeoff between Correctness (declining with larger D_{th}) and Completeness (rising with larger D_{th}), as reported in Table VI. We adopt $D_{th} = 20$ as the operating point, since for emergency planning a phantom shortcut can divert assets into impassable terrain whereas a residual gap merely forces a small detour. Finally, the endpoint-pairing algorithm achieves a 250× speedup over a conventional iterative-morphology + brute-force implementation (see Table VII), via Zhang-Suen thinning through cv2.ximgproc.thinning and KD-Tree spatial indexing, ensuring the refinement does not bottleneck the overall response.

Runtime is dominated by Zhang-Suen thinning and KD-Tree construction, both of which are independent of D_{th}. Varying $D_{th} \in \{20, 30, 40\}$ produced no measurable difference at the reported precision (worst case 4.7 s in all three settings).

E. Carrier-Aware Path Planning Experiments

This section validates the end-to-end pipeline on a simulated emergency rescue scenario. We adopt a representative regional task. A rescue mission originates in Hongya County (start, green marker) heading to a disaster site in Ya'an City (goal, blue marker). The distance and topographic complexity, which include a meandering river, forest belts, and a mountain ridge, exercise the full range of behaviors the framework is designed to handle.

Two experimental settings are compared throughout this section.

Experiment 1 (GT Road Network): Planning runs on the unified cost map constructed from the GT road labels of STRoadSet fused with the NX-SegNet land-cover output. This isolates the planning component from upstream perception error and serves as a planning-side upper bound reference.

Experiment 2 (Inferred Road Network): Planning runs on the unified cost map constructed from the HC-RoadNet output (refined via Section III-D) fused with the NX-SegNet land-cover output. This is the operational deployment scenario.

The contrast between these two experiments constitutes a path-level quantitative validation, since it directly measures the impact of perception error on routing performance while holding all other variables constant. We report results in three optimization modes, namely time-priority [$\alpha = 1, \beta = 0$ in (19)], energy-priority ($\alpha = 0, \beta = 1$), and balanced ($\alpha = \beta = 1$).

1) Carrier Profiles and Parameterization: Three heterogeneous emergency carriers are simulated, each with distinct mobility characteristics.

Carrier A is a wheeled logistics vehicle with strong dependence on roads. It is impassable on water bodies and incurs medium-to-high impedance on unstructured terrain due to traction loss.

Carrier B is an amphibious hovercraft that treats water bodies and permanent snow/ice as frictionless corridors due to its air-cushion suspension. It is impassable on dense vegetation (forest and shrubland), which risks tearing the air skirt.

Carrier C is an on-foot rescue team with maximum terrain flexibility. It can traverse forests and shrublands at moderate impedance and is blocked from water bodies in energy-priority

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TABLE VIII
CARRIER-SPECIFIC COST PARAMETERS (NORMALIZED)

Land-Cover Class	Index	Carrier A (E / V)	Carrier B (E / V)	Carrier C (E / V)
Cultivated Land	0	1.5 / 0.8	1.2 / 1.8	1.0 / 1.0
Forest	1	5.0 / 0.2	∞ / 0	1.8 / 0.7
Grassland	2	1.2 / 0.9	1.1 / 1.9	1.1 / 0.9
Shrubland	3	3.5 / 0.4	∞ / 0	2.5 / 0.6
Wetland	4	8.0 / 0.1	1.5 / 1.5	4.0 / 0.4
Water Bodies	5	∞ / 0	0.01 / 999	∞ / 0.01
Tundra	6	2.5 / 0.6	1.8 / 1.4	2.2 / 0.8
Artificial Surfaces	7	1.0 / 1.0	1.1 / 1.9	0.8 / 1.2
Bareland	8	2.0 / 0.7	2.5 / 1.2	1.2 / 0.9
Perm. Snow/Ice	9	7.0 / 0.15	0.01 / 999	5.0 / 0.3
Road (Inferred)	10	0.01 / 999	1.5 / 1.2	0.01 / 999

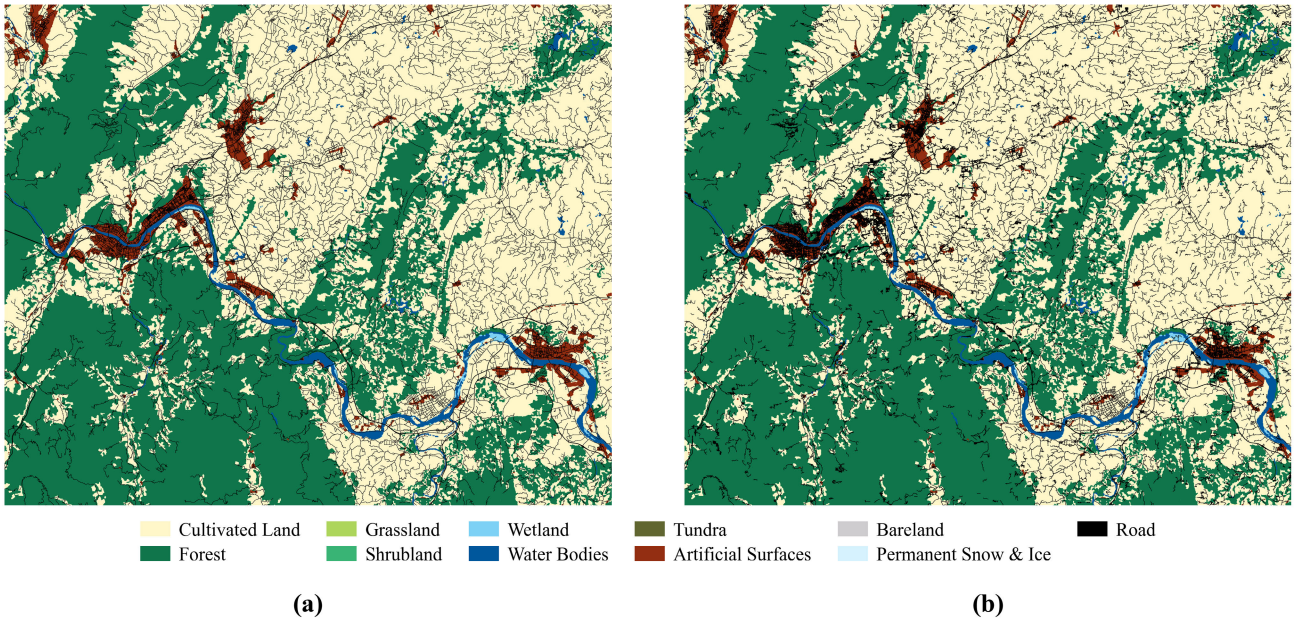

Fig. 10. Unified cost map for Carrier A under the time-priority mode. (a) Experiment 1 (GT). (b) Experiment 2 (Inferred).

and balanced modes ($E = \infty$), with a soft-constraint encoding ($V = 0.01$) that admits emergency wading only under strict time-priority objectives.

Assigning discrete, manually specified impedance weights to land-cover classes is an established practice in GIS-based least-cost-path planning [50], and the platform-dependent nature of terrain trafficability—whereby the same surface imposes different costs on different carriers—is well documented in off-road mobility modeling [51]. Following these conventions, the values in Table VIII encode the relative impedance ordering across classes for each carrier. As emphasized in Section III-E, and consistent with the observation that only relatively optimal paths exist under a given parameterization [50], these constitute relative gradients rather than absolute physical measurements. The carrier-specific energy and speed coefficients are listed in Table VIII.

As emphasized in Section III-E, these values constitute relative impedance gradients calibrated for algorithmic stability and consistent ranking, not absolute physical measurements. We follow two encoding conventions for boundary cases in Table VIII. Preferred terrain uses $E = 0.01$ and $V = 999$, where the small positive e (rather than 0) preserves A* admissibility under the heuristic of (20), as discussed in Section III-E. Impassable terrain is encoded as $E = \infty$, which forces $G = \infty$ in the energy-priority and balanced modes via (19), hard-blocking the planner. The V entry under impassable terrain encodes a finer distinction between two physical regimes. A hard physical constraint (e.g., a wheeled vehicle on water, a hovercraft in dense forest) is encoded as $V = 0$, which yields $G = \infty$ even under time-priority mode and forbids traversal under any optimization objective. A soft constraint that becomes admissible under extreme time pressure (e.g., on-foot wading across shallow water in a

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life-saving emergency) is encoded as $V = 0.01$, which keeps energy-priority and balanced modes blocked but yields a finite, large $1/V$ cost in time-priority mode, allowing the planner to use that pixel only when speed is the sole optimization criterion. This dual encoding lets a single cost table express both inviolable physical limits and emergency-only soft fallbacks.

Before turning to the resulting trajectories, Fig. 10 visualizes the unified cost maps that drive the planner under the two experiments, using Carrier A under time-priority mode as a representative example. The two cost maps share an identical land-cover background derived from NX-SegNet, and they differ exclusively in the road layer overlaid on top of it. In the GT panel (a), the road layer is the GT STRoad-Set label, while in the inferred panel (b), the road layer is the HC-RoadNet output after the four-step refinement of Section III-D. Consequently, any subsequent difference in trajectory between Experiments 1 and 2 must be attributable to the residual fractures and false positives in the inferred road layer, rather than to land-cover perception errors. The cost maps for Carriers B and C and for the energy-priority and balanced modes follow the same GT/Inferred structural pattern and are omitted for brevity.

2) Trajectory-Level Behavioral Analysis: Figs. 11–13 present the generated trajectories under the three optimization modes (time-priority, energy-priority, and balanced, respectively). Each figure is organized as a 3×2 grid in which rows correspond to the three carriers (A–C from top to bottom) and columns correspond to the two experimental settings, namely, the GT road network (Experiment 1, left) and the inferred road network (Experiment 2, right). This layout allows the reader to inspect, at a glance, both the intercarrier behavioral diversity under a fixed optimization mode and the perception-error effect that distinguishes Experiment 2 from Experiment 1 for each carrier.

Three behavioral regimes are observed across the three figures.

Carrier A (Vehicle, Top Row of Each Figure): The trajectories adhere strictly to the road network in both experimental settings. In the GT panels (left column), the path follows a clean, continuous road backbone. In the inferred panels (right column), the path follows the same road corridor but exhibits localized detours wherever the inferred network contains residual fractures, since Carrier A's strict road-dependence forces the planner to either backtrack to an alternate access point or descend briefly into off-road grid search at each gap. The visual difference between Experiments 1 and 2 for Carrier A is therefore the clearest direct illustration of how upstream perception error propagates to the trajectory shape. Across the three modes (see Figs. 11–13), the time-priority and energy-priority paths for Carrier A are visually identical, because the road class is optimal under both metrics for this carrier; the balanced mode produces the same path. This validates the road-priority hierarchical strategy, which correctly collapses to road-graph traversal when the carrier's optimal medium is unambiguous.

Carrier B (Hovercraft, Middle Row of Each Figure): The most diverse behavior across modes, and consequently the row that differs most between Figs. 11–13. In the time-priority figure (see Fig. 11), the planner exploits the river as a high-speed corridor but occasionally cuts across flat terrain to bypass the river's meanders, accepting small friction penalties to reduce total path length. In the energy-priority figure (see Fig. 12), the trajectory adheres rigorously to the river's centerline regardless of curvature, because the water class incurs essentially zero energy cost ($E = 0.01$). The balanced figure (see Fig. 13) yields an intermediate behavior. The contrast between GT and Inferred panels is small for Carrier B in all three modes, because its operation is dominated by the water domain, which is cleanly classified by NX-SegNet and is unaffected by inferred-road fragmentation. This contrast across modes demonstrates that the same cost map produces qualitatively different optima under different $\alpha : \beta$ weightings, confirming that the optimization objective genuinely steers behavior rather than acting as a normalization knob.

Carrier C (On-Foot, Bottom Row of Each Figure): Despite the carrier's ability to traverse forests and shrublands, the planner consistently routes via roads when available, because the road impedance is far lower than vegetation impedance even for an unencumbered pedestrian. The GT and inferred panels are visually similar across all three modes, with the inferred panels exhibiting slightly more frequent short off-road segments where the inferred road network is locally fragmented. These off-road segments traverse low-impedance vegetation rather than producing the high-cost detours seen in Carrier A's row, which is why Carrier C is the least affected by perception error in the runtime data of Section IV-E3. This is the desired behavior, namely, that the off-road capability is reserved as a backup rather than a default.

3) Path-Level Quantitative Comparison (GT Versus Inferred Road Network): Table IX reports the planning runtime under both experiments across all carrier-mode combinations. This is the central quantitative validation of the routing system, and it complements the visual comparison provided in Figs. 11–13 by quantifying how perception error propagates to path-level performance.

Three findings emerge from Table IX. First, in the GT setting (Experiment 1), Carriers A and C, which are both road-dependent, plan in tens of seconds across all modes, because the road-priority hierarchical strategy collapses the long-range search into sparse road subgraph traversal. Carrier B in time-priority mode is the slowest GT case (3236 s) because its preferred class (water) is large, contiguous, and lacks the linear topology that the hierarchical strategy can exploit, and the planner therefore performs a true area search over the water domain. Note, however, that the balanced and energy-priority modes for Carrier B drop sharply (165–217 s), demonstrating that suitable cost-gradient parameterization (see Section III-E) can prune the search space even for unstructured-medium carriers.

Second, comparing Inferred to GT across all entries, perception error propagates nonuniformly to runtime. Carrier C is least affected (1.4× to 3× slowdown), because its high terrain flexibility allows it to gracefully bypass small road fractures via off-road detours. Carrier B is moderately affected, because its operation is dominated by water (cleanly

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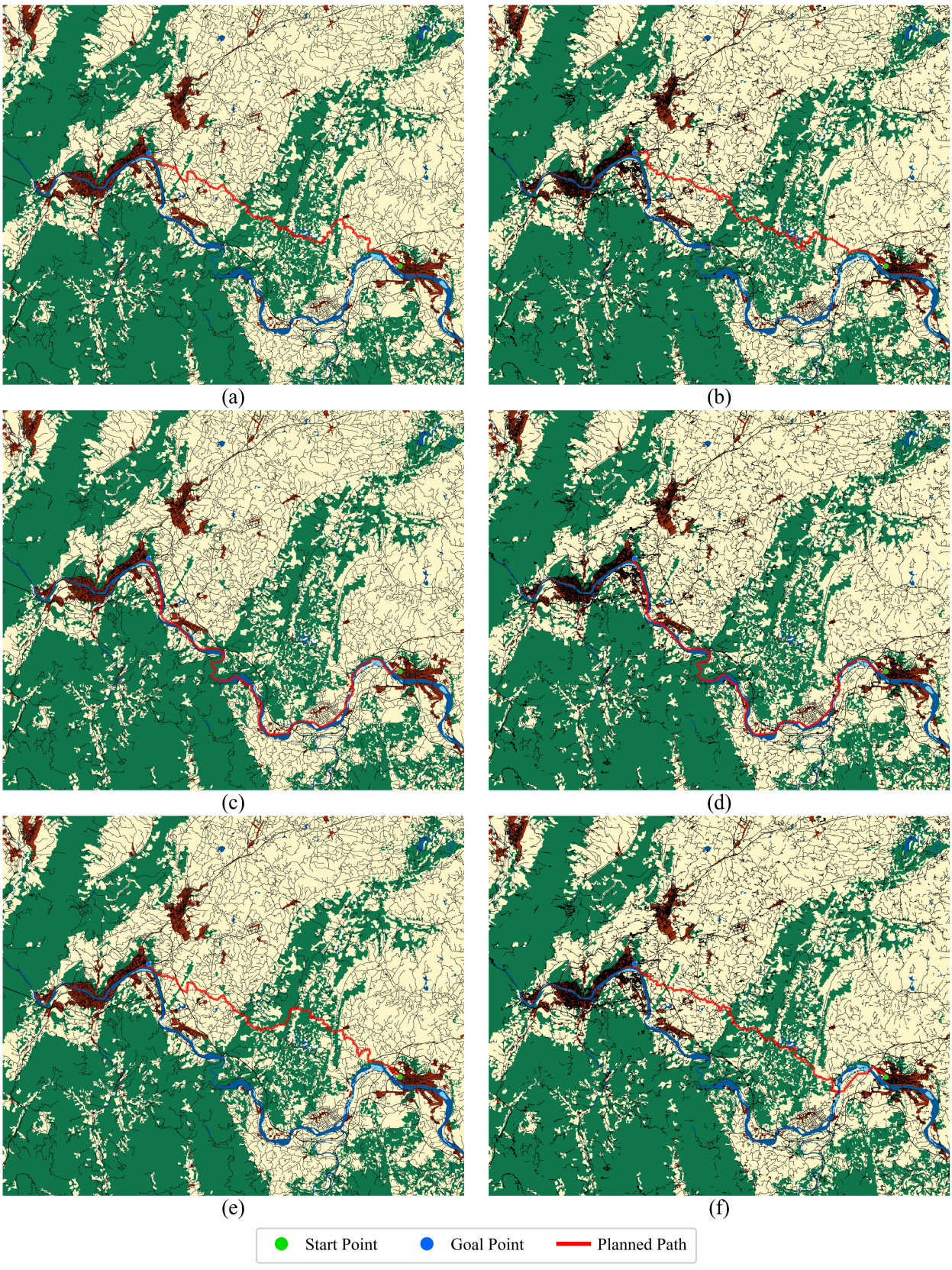

Fig. 11. Generated trajectories under time-priority mode ($\alpha = 1$ and $\beta = 0$). This figure is a 3×2 grid in which rows correspond to Carrier A [(top) vehicle], Carrier B [(middle) hovercraft], and Carrier C [(bottom) on-foot]. The left column shows Experiment 1 (GT road network from STRoadSet labels); the right column shows Experiment 2 (inferred road network from HC-RoadNet after refinement). (a)–(f) Follow the order A-GT, A-Inferred, B-GT, B-Inferred, C-GT, and C-Inferred, respectively. Green markers denote start, blue markers denote goal, and red curves are the planned trajectories.

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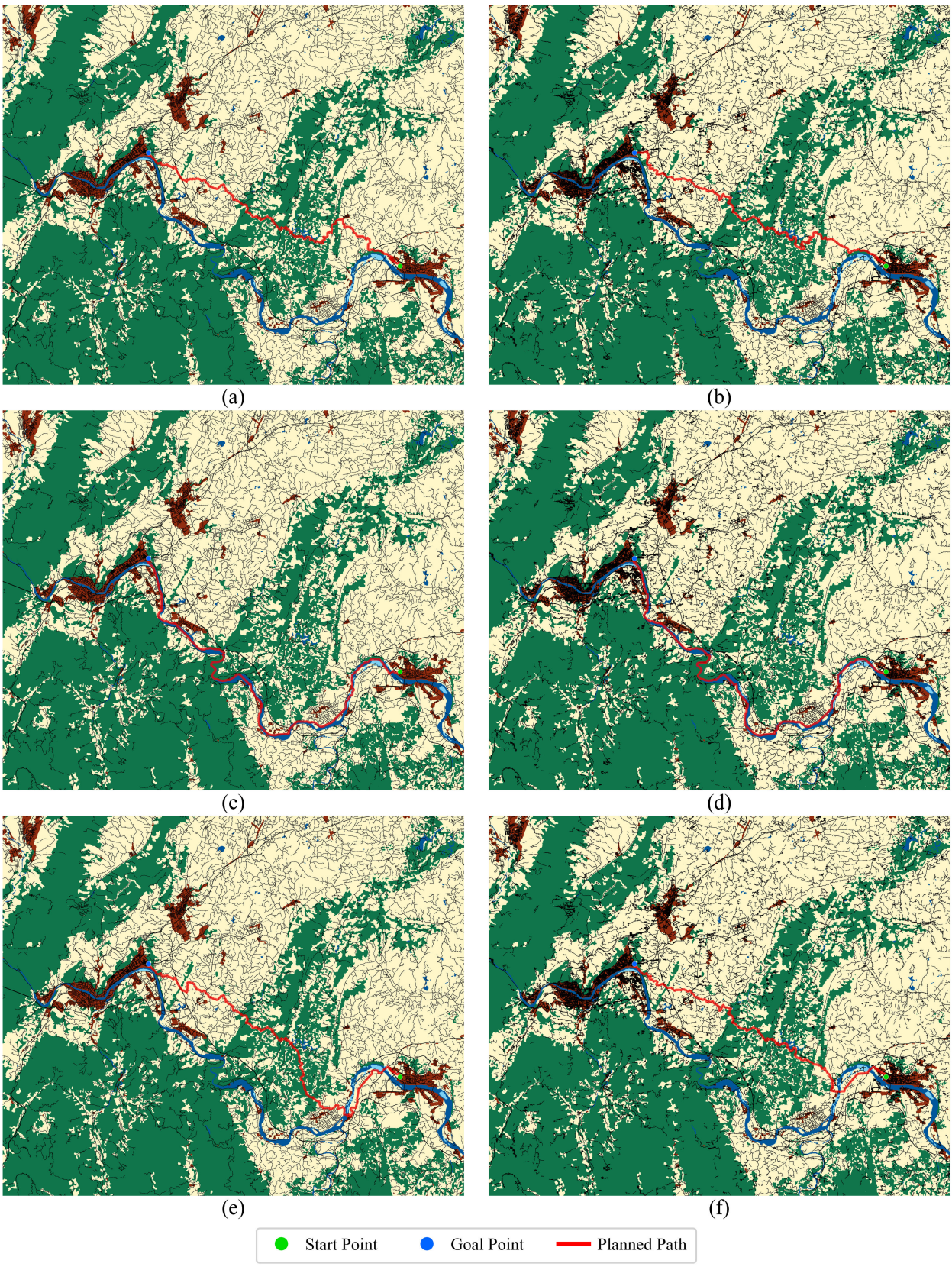

Fig. 12. Generated trajectories under energy-priority mode ($\alpha = 0$ and $\beta = 1$). This figure adopts the same 3×2 layout. (a)–(f) Follow the order A-GT, A-Inferred, B-GT, B-Inferred, C-GT, and C-Inferred, respectively.

classified by NX-SegNet) rather than roads. Carrier A is most affected, with up to 15× slowdown in the balanced mode (63.72 → 987.57 s). This is a direct consequence of the strict road-dependence. Every residual fracture in the inferred road

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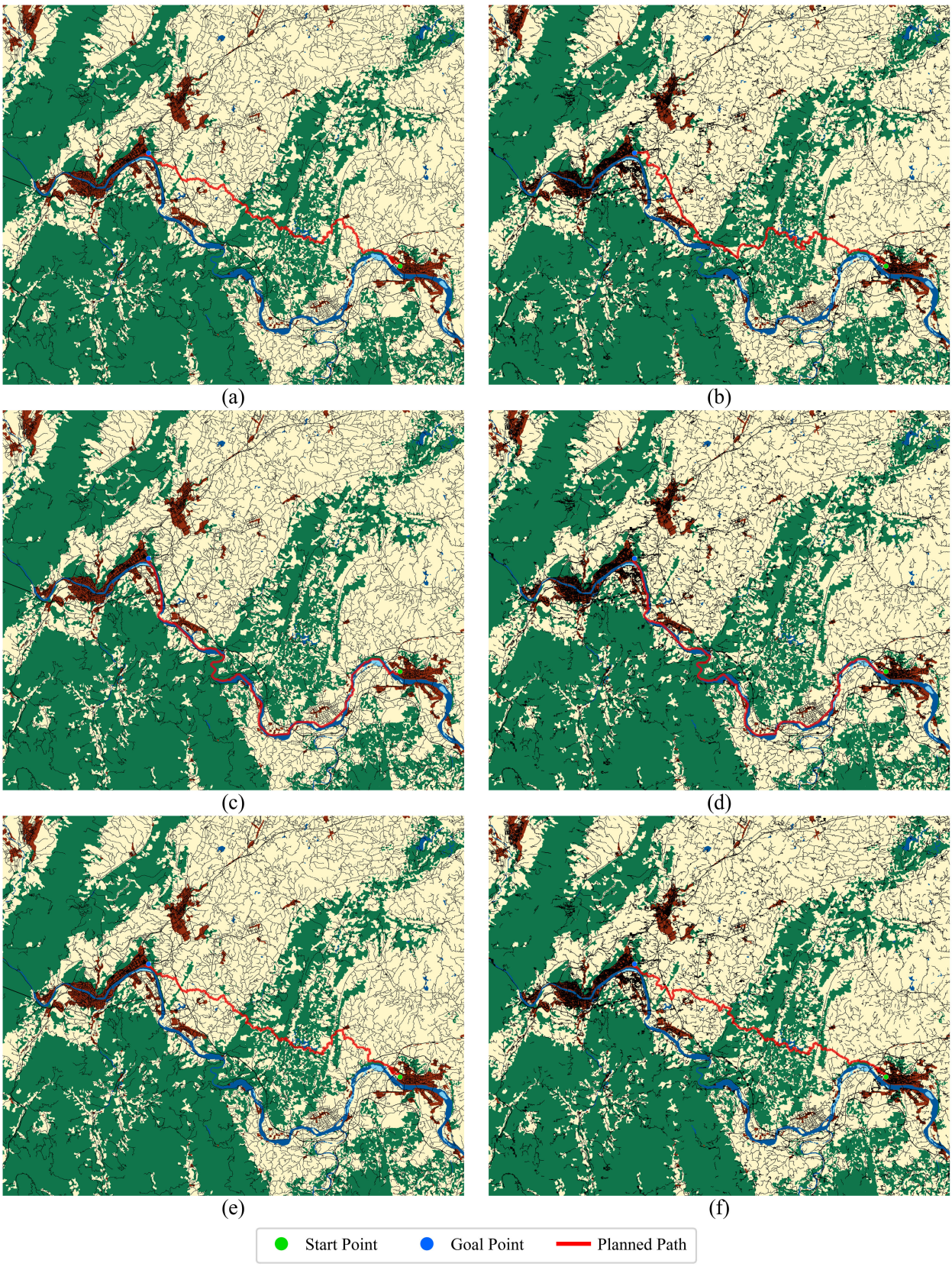

Fig. 13. Generated trajectories under balanced mode ($\alpha = \beta = 1$). This figure adopts the same 3×2 layout. (a)–(f) Follow the order A-GT, A-Inferred, B-GT, B-Inferred, C-GT, and C-Inferred, respectively.

network forces the planner to either: 1) backtrack and seek an alternate access point or 2) descend into expensive grid-level search across high-cost off-road pixels. The balanced mode is the worst case because both speed and energy gradients agree

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TABLE IX
PATH PLANNING RUNTIME COMPARISON: INFERRED VERSUS GT ROAD NETWORK (UNIT: SECONDS)

Mode	Carrier A (GT)	Carrier A (Inferred)	Carrier B (GT)	Carrier B (Inferred)	Carrier C (GT)	Carrier C (Inferred)
Time-priority	41.88	171.21	3236.41	4825.71	12.82	38.99
Energy-priority	41.08	238.87	217.30	489.48	43.94	65.79
Balanced	63.72	987.57	165.79	222.64	44.48	61.36

that off-road exploration is undesirable, leaving the planner few escape hatches.

Third, despite the runtime increase, the planner reliably finds a feasible path in every inferred case, with no failures observed. This robustness owes to the unified cost map's design. Even when the inferred road subgraph is disconnected, the land-cover layer provides traversable bridges, and the system's fallback to direct full-grid A* (Section III-F2) ensures path existence whenever the underlying terrain is feasible. The price of perception error is paid in runtime, not in route validity, which is a meaningful distinction in operational deployment.

These results substantiate two claims that have been raised as concerns in the literature on remote-sensing-driven routing. First, quantitative path-level evaluation, beyond purely qualitative trajectory inspection, is feasible and informative when the GT road network is treated as a planning-side upper bound. Second, the dominant operational risk of imperfect perception is computational, not topological, given a well-designed cost map.

V. DISCUSSION

Although the proposed framework demonstrates promising results in simulated emergency scenarios, several limitations warrant discussion and indicate directions for future research.

First, while HC-RoadNet combined with the four-step refinement pipeline (see Section III-D) substantially improves the topological completeness of the road layer, the inferred network remains imperfect in extremely complex environments. Dense ridge shadows, persistent cloud cover, and severely degraded postdisaster scenes can produce extended fractures that exceed the operating range of the endpoint-pairing module. Although the unified cost map mitigates these residual gaps via off-road traversal across low-impedance land-cover, such pseudo-gaps still inflate planning runtime, as quantified in Table IX. Future work could embed geometric topology constraints into the loss function, or integrate centerline-tracking algorithms with multi-temporal imagery to further enhance connectivity.

Second, the current carrier-aware cost model employs relative impedance gradients calibrated for algorithmic stability rather than absolute physical measurements. A formal sensitivity analysis of these parameters, together with empirical calibration against vehicle test data or expert-designed routes, is left for future work. Moreover, the model simplifies terrain geometry by assuming a flat-earth approximation within each pixel, neglecting slope and elevation. In the mountainous Sichuan-Tibet region, a steep paved road may consume more energy than a flat grassland path. Integrating high-resolution digital elevation model data into cost-map generation will be a priority in our next research phase to enable 3-D terrain-aware routing.

Finally, this study focuses on static routing based on a single postdisaster image. Disaster zones are dynamic environments in which risks evolve rapidly, and a static path may become hazardous shortly after planning. Future research will extend the framework to a dynamic replanning system that updates the semantic cost map as new imagery arrives, allowing rescue routes to adapt continuously throughout a mission.

VI. CONCLUSION

This article has presented an end-to-end framework that turns a single postdisaster remote sensing image into a carrier-aware navigable route. Rather than pursuing a single best-performing perception model, we argue for and empirically validate a different design principle: in the emergency perception-to-decision chain, perception errors should be made recoverable rather than eliminated. HC-RoadNet deliberately operates in a high-recall, wide-boundary regime; the four-step refinement pipeline absorbs the resulting false positives and reconstructs broken topology using a land-cover prior; and the carrier-aware cost map allows the planner to bridge any residual gaps via traversable terrain. The path-level GT-versus-Inferred experiments confirm that, under this design, the price of imperfect perception is paid in planning runtime rather than in route feasibility, a tradeoff that aligns naturally with the operational priorities of disaster response.

Two deliverables of this work, the STRoadSet and STLandSet datasets together with the public release of the full source code, are intended to lower the entry barrier for follow-up research in topographically complex, disaster-prone regions, which remain underrepresented in the existing remote sensing literature.

The framework is not yet complete. Slope, surface roughness, and time-varying hazards are absent from the current cost model; carrier parameters are calibrated as relative impedance gradients rather than from physical measurements; and routing remains static. Addressing these limitations, particularly through the DEM integration and dynamic replanning over streaming imagery, defines the next phase of this research and a productive direction for the broader field.

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